

Department of Chemical Engineering

Final-Year Project (C) 478

Bayesian PCA With Automatic Dimensionality Selection For Process Monitoring And Fault Detection.

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Abstract

Principal component analysis (PCA) is a dimensionality reduction technique that enhances the interpretation of complex data in the presence of noise. By retaining certain principal components (PCs) and discarding the others, a PCA model can be used to separate the features and noise into separate subspaces. The aim of this research is to compare PCA to Bayesian PCA (BPCA) in the context of process monitoring and fault detection.

A major challenge in PCA lies in the appropriate choice of retained PCs, which falls under the broader model complexity problem. This is a significant challenge because complex datasets often do not have an obvious or easily determined latent dimensionality, and this choice underpins the PCA model's performance. BPCA may assist with the model complexity problem due to automatic relevance determination (ARD), whereby the importance of PCs is learned directly from the data. BPCA is a formulation of PCA that uses prior distributions over the latent and data variables. An expectation-maximisation (EM) algorithm is used to learn the model parameters: weights, variance, and hyperparameters that determine the model complexity.

PCA and BPCA can be used for monitoring by evaluating statistics that evaluate the subspace projections of new data compared to the training data. These statistics are compared to confidence thresholds to detect abnormal process operation. To best compare the two techniques, they should be applied to the same case study imitating real industrial process monitoring.

A case study was performed by using MATLAB Simulink™ to simulate a jacketed continuously stirred tank reactor (CSTR) to generate artificial plant data for testing fault detection. To emulate realistic plant data, measurement noise and random-walk process variation is implemented. Faul types such as inlet flow rate step changes, pipeline pressure step changes, catalyst deactivation, fouling, and reactor temperature setpoint step changes are considered. Process monitoring is achieved by evaluating the subspace projections of new data compared to the training data. These statistics are compared to confidence thresholds to detect abnormal process operation.

PCA's fault detection performance reflected reported literature values and reaffirmed the practice of implementing statistics two monitor the two (2) subspaces separately. BPCA did not prune any components during the EM algorithm, so PC selection was performed by using an analogy to the most common method used for PCA. BPCA produced inexplicably large T^2 values, the cause of which being the EM algorithm was ruled out by validating the weights against PCA and probabilistic PCA results and finding that they concur. Additionally, the Q-statistic and latent variables were within expected bounds. Therefore, the only suspicious term remaining is that of the variance of the posterior latent variable.

Given the unexplained BPCA results and no native PC pruning, BPCA is not recommended for use over PCA for process monitoring and fault detection. However, the reasons behind the T^2 results being so large is worth investigating for the application of BPCA where it is computationally advantageous to PCA (such as very large industrial datasets).

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Nomenclature

Symbols		
A	Cross-sectional area	dm^2
A_C	Heat transfer area	dm^2
c	The normal deviate	
C	The covariance matrix for principal component analysis	
CONT	T^2 statistic contribution by a variable for an observation	
C_A	Concentration of reagent A in the reactor	mol L^{-1}
C_{AF}	Concentration of reagent A in the feed to the reactor	mol L^{-1}
C_p	Heat capacity of the liquid in the reactor	J kg^{-1} K^{-1}
C_{pC}	Heat capacity of the coolant	J kg^{-1} K^{-1}
C_v	Valve coefficient	
diag	Matrix diagonal operator; creates a diagonal matrix from a vector and vice-versa.	
D	The total number of variables observed	
e	The vector of prediction errors for each variable	
E	Activation energy	J mol
$\mathbb{E}$	Expectation operator	
$\in$	Element in	
f(l)	Valve characteristic	
F()	The F distribution	
g_s	Specific gravity of liquid	
h	Liquid level in the reactor	dm
h_0	The factored-out Q statistic distribution sum term	
ΔH	Heat of reaction	J mol^{-1}
I	Identity matrix	
k	The number of retained principal components	
k_0	Pre-exponential coefficient	min^{-1}
K	The covariance matrix for probabilistic PCA	
l	Fraction valve opening	
lim	Significance level for confidence limits	
ln	Natural logarithm	
M	Matrix used for the matrix inversion identity of the covariance matrix	
N	The total number of observations	

N_p	The candidate pool	
N_{CF}	Valve equation unit conversion factor	L min^{-1} $Pa^{-0.5}$
N_{DB}	The total number of data sets that were evaluated that are actually similar to the snapshot	
N_1	The number of data sets in the candidate pool that are actually similar to the snapshot data set	
N_2	The number of data sets in the candidate pool that are not actually similar to the snapshot data set	
$\mathcal{N}()$	The Normal distribution	
o	A single observed element of data	
$\underline{\bar{\boldsymbol{o}}}$	Vector containing the mean value of each variable in an observed dataset	
$\underline{\boldsymbol{O}}$	Matrix of observed data	
p	The pool accuracy	%
p	An eigenvector from the retained principal components matrix	
$\underline{\mathbf{P}}$	The matrix of retained principal components	
P()	Probability	
Q	Reactor outlet flow rate	L min^{-1}
Q_F	Reactor inlet flow rate	L min^{-1}
Q_c	Coolant flow rate	L min^{-1}
r	The limit for the number of times a statistic violates its confidence limit	
R	Universal gas constant	J mol^{-1} K^{-1}
$\mathbb{R}$	All real numbers	
$\underline{\mathbf{S}}$	The diagonal matrix of singular values	
S	An individual singular value	
t	Time	s
t	A single PCA score	
$\underline{\mathbf{t}}$	Vector of the PCA scores	
$\underline{\mathbf{T}}$	The matrix containing the vectors of latent variables for every observation	
Tr	Trace operator	
T^2	Hotelling's T^2 statistic	
T	Reactor (outlet) temperature	K
T_C	Coolant (outlet) temperature	K
T_{CF}	Coolant inlet temperature	K
T_F	Reactor feed temperature	K
U	Heat transfer coefficient	J min^{-1} K^{-1} dm^{-2}
$\underline{\mathbf{U}}$	The left unitary matrix	

$\underline{\mathbf{v}}$	An eigenvector	
var	Variance operator	
V	Volume of the liquid in the reactor	L
$\underline{\mathbf{V}}$	The right unitary matrix, the matrix of eigenvectors	
V_C	Volume of the coolant in the cooling jacket	L
$\underline{\mathbf{w}}$	A vector of corresponding to a single principal component in the weight matrix	
$\underline{\mathbf{W}}$	Bayesian principal component analysis weights matrix	
x	A single observation of one variable from a normalised dataset	
$\underline{\boldsymbol{x}}$	Row vector of a single observation of several variables from a normalised dataset	
$\underline{\bar{\mathbf{x}}}$	The row vector containing the mean of each variable from a normalised dataset	
$\underline{\mathbf{X}}$	A matrix of normalised data	
$\mathcal{X}$	Chi-squared distribution	
$\underline{\mathbf{z}}$	Vector of a single observation of each latent variable	
$\underline{\mathbf{Z}}$	Matrix of latent variables	
Greek symbols		
α	A single hyperparameter	
$\underline{\boldsymbol{\alpha}}$	The vector of hyperparameters for the Bayesian PCA model	
ξ	Overall effectiveness	
$\underline{\boldsymbol{\epsilon}}$	The vector noise term in the probabilistic PCA formulation	
λ	An eigenvalue	
$\underline{\boldsymbol{\lambda}}$	The diagonal matrix of eigenvalues	
$\underline{\boldsymbol{\mu}}$	The vector containing the mean for each observed variable	
η	The pattern matching efficiency	
$\omega(\beta)$	The optimal hard threshold coefficient	
Φ	The cumulative normal distribution	
ρ	Density of the reactor liquid	
ρ_C	Density of the coolant	
σ	The standard deviation of the Gaussian noise for the noise term of probabilistic PCA	
$\hat{\tau}_*$	Optimal hard threshold for singular values from a non-square observation matrix with an unknown noise parameter	
θ	Q statistic distribution definition sum term	
Subscripts and superscripts		
H	Corresponding to the historical dataset	

j	Denoting a value corresponding to any individual variable from an observation
k	Indicating only the portion of the matrix corresponding to the retained principal components
lim	Indicating a value corresponding to a significance level for confidence limits
m	The mth of a quantity relating to principal components
med	The median value
M	Measurement
n	The nth observation
part	Variable with this superscript is defined by part of the host-name variable's definition
Q	Of the reactor outlet valve
QC	Of the coolant inlet valve
S	Corresponding to the snapshot dataset
SS	Steady-state value
T	Transpose
1, 2, 3	Indicates the first, second, and third respective sum terms of the Q statistic distribution.
Acronyms	
ARD	Automatic relevance determination
BPCA	Bayesian principal component analysis
CSTR	Continuously stirred tank reactor
EM	Expectation maximisation
FBFPCA	Fast Bayesian functional principal component analysis
FDI	Fault detection and identification
GMM	Gaussian mixture model
MSPC	Multivariate statistical process control
OHT	Optimal hard threshold
PC	Principal component
PCA	Principal component analysis
SPE	Squared prediction error
SuSiE	Sum of single effects
VBPCA	Variational Bayesian functional principal component analysis
WWTP	Wastewater treatment plant

1 Introduction

Industrial processes need control to operate safely and economically. It is impossible for any plant to operate at steady state endlessly due to the inevitable nature of faults. Faults can be defined as the interruptions to desired process operation. They can be the result of unpredictable phenomena, the consequence of the operation of connected process units, or simply plant startup and shutdown. Diagnosing the cause of faults allows for better planning for and prevention of future occurrences. Fault detection and identification (FDI) is paramount towards this end.

Process control functions on the principal of moving variability from one process variable to another. To effect this, controllers adjust final elements; valve positions are actuated, pump or motor speeds are modulated, and heater or furnace temperatures are modified. However, controllers need information about the process before they can effect change. The typical control algorithm uses digital measurements from sensors in the process(es), which are often stored by the plant's data historian. It is wasteful to not analyse this valuable data of the plant's operation to glean further insight regarding the process(es)'s performance.

However, all data collected are subject to noise; unpredictable and uncontrollable errors in measurement. The relative magnitude of this uncertainty with respect to the true value influences how we can interpret the data. Statistical techniques provide ways in which data can be more reliably analysed and often rely on statistics about the data. Statistics, such as the mean and standard deviation, help to characterise the dataset for comparison to other datasets of the same type of measurements.

Where each aspect of an object that is measured is a dimension of the dataset, more complex objects require higher dimensional datasets to fully describe their state. However, most humans naturally have a poor ability to interpret high-dimensional data and are largely limited to visualising data through two- or three-dimensional graphs. This makes it challenging to assess the underlying and unmeasurable relationships hidden in data, which are termed latent variables. Therefore, statistical techniques, such as principal component analysis (PCA) that can reduce the dimensionality of a dataset by attempting to extract the most significant contributions (or 'features') to a latent variable are valuable.

However, the chosen level of model complexity poses a new challenge. Model complexity relates the fit of the model to the training data. A more complex model fits training data better but may perform worse for new data. Therefore, it is important to make informed decisions to maximise the model's performance over new and training data. This becomes especially challenging for datasets with a large amount of recorded variables where there is no clear motivation for the choice of model complexity. A more advanced probabilistic formulation of PCA, Bayesian PCA (BPCA), attempts to resolve this issue through automatic relevance determination (ARD): using the training data to determine the resulting model's complexity.

The aim of this research is to evaluate BPCA for FDI applied to the operation of a simulated jacketed non-isothermal continuously stirred tank reactor (CSTR).

2 Objectives and Key Questions

2.1 Key Questions

Towards the aim of this research to assess BPCA, a reference point is needed. The most appropriate choice for a reference is PCA as: it has a long history of use for FDI, is widely applied, and is one of the foundations of BPCA. In comparing the two, one consequently asks

- "does BPCA offer any notable improvements over PCA for FDI?"

However, PCA does not benefit from the assistance in choosing model complexity that is provided by ARD in BPCA. Thus, another valuable question is

- "how do methods for selecting the number of retained principal components (PCs) for PCA compare to the results of ARD from BPCA?"

2.2 Objectives

The research's objectives serve to answer the key questions. Naturally, the first objective must be

- to implement and compare the performance of PCA and BPCA for FDI through commonly used pattern-matching statistics.

Similarly, the next objective becomes

- to investigate techniques for PC selection for PCA and compare their result to that of BPCA's ARD.

It is important for the primary objectives' investigation to use data that is of comparable complexity to real plant data. This allows the research to communicate meaningful results to professionals who employ, or are interested in employing, the same techniques in an industrial setting. Therefore, a process model used in published PCA pattern-matching research should be used, and the first objective is

- to create the CSTR model from Johannesmeyer (1999) with autoregressive noise applied to process inputs and white noise on sensor measurements, with the respective noises' variance as specified by Johannesmeyer and Singhal (2002) and Wong (2006).

Additionally, due to the application of random noise, a representative sample of both normal operation and fault occurrences should be generated. This is to reduce the risk of unlikely scenarios from being over-represented in the data produced. As it is difficult to motivate what a 'representative' sample size is, published research should be used as a guideline. Thus, the second objective must be

- to create a data-generation protocol that is similar to that of Johannesmeyer and Singhal (2002) and Wong (2006).

As one of the core aspects of this research, the choice of sensible fault types and magnitudes must be considered carefully to generate results from which an industrially relevant conclusion can be drawn. Once more, the author must turn to published literature to motivate their choices. As the third objective, this research must

- define fault types and magnitudes based on Johannesmeyer (1999), Johannesmeyer and Singhal (2002), and Wong (2006).

3 Literature Review

A Note on Conventions and Nomenclature

Due to the broad application of techniques discussed within different fields, researchers often adopt different conventions. For this reason, the author finds it necessary to be explicit about their nomenclature and conventions. As far as possible, the author tries to use existing conventions while allowing their nomenclature to remain unchanged for the coding implementation of relevant theory.

Matrix multiplication is not explicitly shown but can be assumed to be the default in this text wherever matrices are multiplied. Element-wise multiplication, the Hadamard product, is explicitly denoted by use of '$\odot$'. Matrices will be treated as consisting of column vectors for the desired outcome of matrix multiplication. Naturally, all vectors are treated as column vectors. Matrices will be both in bold and underlined, as some characters are not noticeably different when bold versus not.

3.1 Principal Component Analysis

3.1.1 Model development

Excellent texts that thoroughly break down PCA and its different forms are provided by Bishop (2006); Brunton and Kutz (2022); Russel, Chiang, and Braatz (2000); and Ge and Song (2013a), who are all heavily relied upon for this section.

Data, such as several recorded measurements, can be presented as a matrix by

$$\underline{\mathbf{O}} = \begin{bmatrix} o_{1,1} & \cdots & o_{1,\mathrm{D}} \\ \vdots & \ddots & \vdots \\ o_{\mathrm{N},1} & \cdots & o_{\mathrm{N,D}} \end{bmatrix}. \qquad \textbf{(1)}$$

whose N rows represent observations, with each of the D columns as a distinct measured variable. Therefore, a single row vector given as

$$\underline{\overline{\mathbf{o}}} = \frac{1}{\mathrm{N}} \sum_{n=1}^{\mathrm{N}} \boldsymbol{o}_n \qquad \textit{(2)}$$

can represent the mean values for each of the variables across all observations, summed across the columns in each of the N rows $\underline{\boldsymbol{o}}_{\mathrm{n}}$. Using these definitions, the observed data's variance is written as

$$\mathrm{var}(\underline{\mathbf{O}}) = \frac{1}{\mathrm{N}-1} \sum_{n=1}^{\mathrm{N}} (\boldsymbol{o}_n - \underline{\overline{\mathbf{o}}})^2, \qquad \textit{(3)}$$

where it is implied that the mean $\underline{\overline{\mathbf{o}}}$ is subtracted from each row. For PCA to have the best effect, normalised data $\underline{\mathbf{X}}$ are needed (Brunton & Kutz, 2022: 28) and given by

$$\underline{\mathbf{X}} = \frac{\underline{\mathbf{O}} - \underline{\overline{\mathbf{o}}}}{\mathrm{var}(\underline{\mathbf{O}})}, \qquad \textit{(4)}$$

meaning the observed data are made to have a mean of zero and variance of unity (1). As before, the normalised data consists of row vectors ($\underline{\boldsymbol{x}}$) and with an overall mean ($\underline{\overline{\boldsymbol{x}}}$) as defined by equation (2). For the scaled data populated by row convention, the sample covariance matrix is defined as

$$\underline{\boldsymbol{C}} = \frac{\underline{\mathbf{X}}\underline{\mathbf{X}}^{\mathrm{T}}}{N-1} \qquad \textit{(5)}$$

PCA can be derived as the eigenvector ($\underline{\mathbf{v}}$) decomposition of this covariance matrix for each (m) eigenvalue (λ) by

$$\underline{\boldsymbol{C}}\underline{\boldsymbol{v}}_m = \lambda_m \underline{\boldsymbol{v}}_m. \qquad \textit{(6)}$$

Therefore,

$$\underline{\boldsymbol{C}} = \underline{\boldsymbol{V}}\underline{\boldsymbol{\lambda}}\underline{\boldsymbol{V}}^T \qquad \textit{(7)}$$

using the matrix of eigenvectors ($\underline{\boldsymbol{V}}$) termed 'principal components' (PCs) and the matrix of eigenvalues ($\underline{\boldsymbol{\lambda}}$) (Russell *et al.*, 2000; Bishop, 2006). Each eigenvalue is the variance in the data that is associated with the corresponding principal component. The alternate, and most commonly applied form of PCA, is done by performing singular value decomposition (SVD)

$$\underline{\boldsymbol{X}} = \underline{\boldsymbol{U}}\underline{\boldsymbol{S}}\underline{\boldsymbol{V}}^{\boldsymbol{T}}, \qquad \textit{(8)}$$

which yields the matrices of left ($\underline{\mathbf{U}}$) and right ($\underline{\mathbf{V}}$) transposed (superscript 'T') singular vectors and the diagonal matrix of singular values ($\underline{\mathbf{S}}$). SVD is computationally intense and a large portion of the left and right singular matrices. Specifically, the . The 'economy' SVD can instead be performed, where only of the entries in the non-zero partition. Where the data consists of row vectors and the number of observations N is much larger than the number of variables D, the right singular vector matrix's

columns contain the eigenvectors of the data's covariance matrix. The diagonal matrices of singular values and eigenvalues ($\underline{\boldsymbol{\lambda}}$) are related by (Brunton & Kutz, 2022).

$$\underline{\boldsymbol{\lambda}} = \frac{\underline{\boldsymbol{S}}^{\mathrm{T}}\underline{\boldsymbol{S}}}{N-1}. \tag{9}$$

The original data are projected onto the chosen k number of retained PCs ($\underline{\boldsymbol{P}}$) to produce the PC scores ($\underline{\boldsymbol{T}}$)

$$\underline{\boldsymbol{T}} = \underline{\boldsymbol{P}}^{\mathrm{T}}\underline{\boldsymbol{X}}, \tag{10}$$

where k is smaller than M. This matrix now contains the latent variables captured by the PCA model and is what will be analysed for fault detection.

3.1.2 Model complexity

One method for determining the number of PCs to retain is using a Scree plot to manually find an 'elbow' in the line. A Scree plot is constructed by plotting the value of each eigenvalue ordered from right to left as largest to smallest, with an example given by Figure 1. Typically, an 'elbow' in the eigenvalues is used to choose the principal components to retain. An elbow refers to a sharp drop in value between eigenvalues that suggests a border between the feature and noise space. Applying this approach to Figure 1, one would motivate retaining the first two principal components. A second, smaller, less-defined elbow exists between principal components 8 and 10.

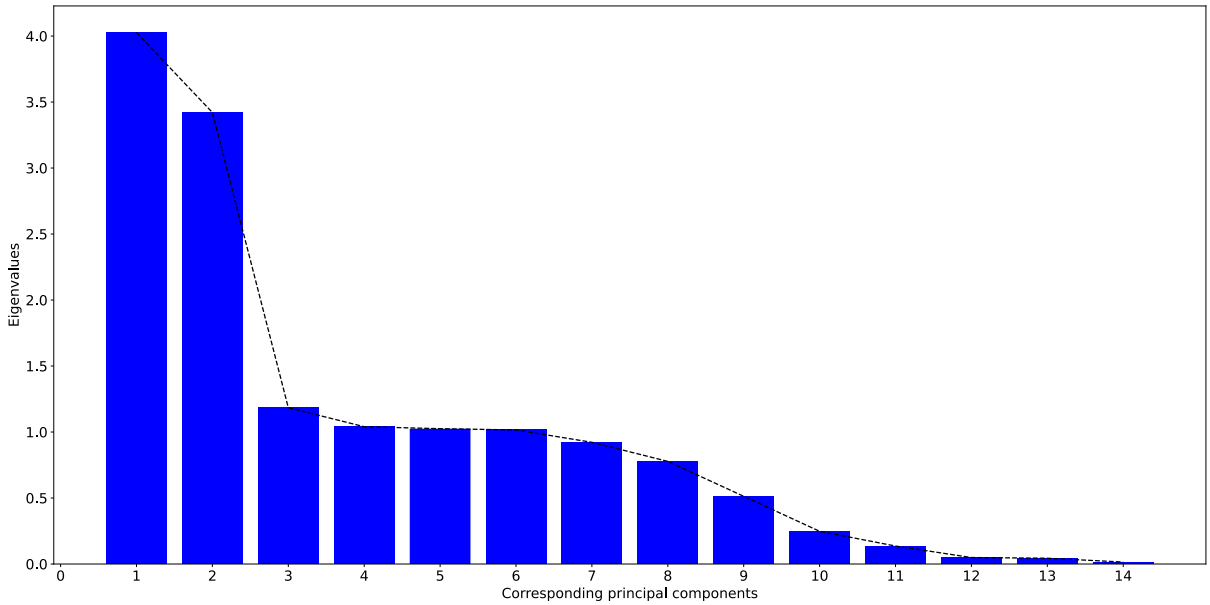


Figure 1. A Scree plot that plots the eigenvalues for each principal component in order from largest to smallest. PCA was performed on the results from simulating a CSTR at normal operating conditions with control as well as process and measurement noise.

Another common method is based on the cumulative explained variance (CEV) by dividing the sum of each eigenvalue up to a principal component by the sum of all eigenvalues. This uses the property that

each eigenvalue is the variance of the data explained by the corresponding principal component. By choosing a cut-off limit, the number of retained components is based on the number of respective eigenvalues required to meet this limit, starting from the largest and adding each successive eigenvalue until the limit is exceeded. The cut-off limit is typically 95 %. An example of applying this method is shown in Figure 2.

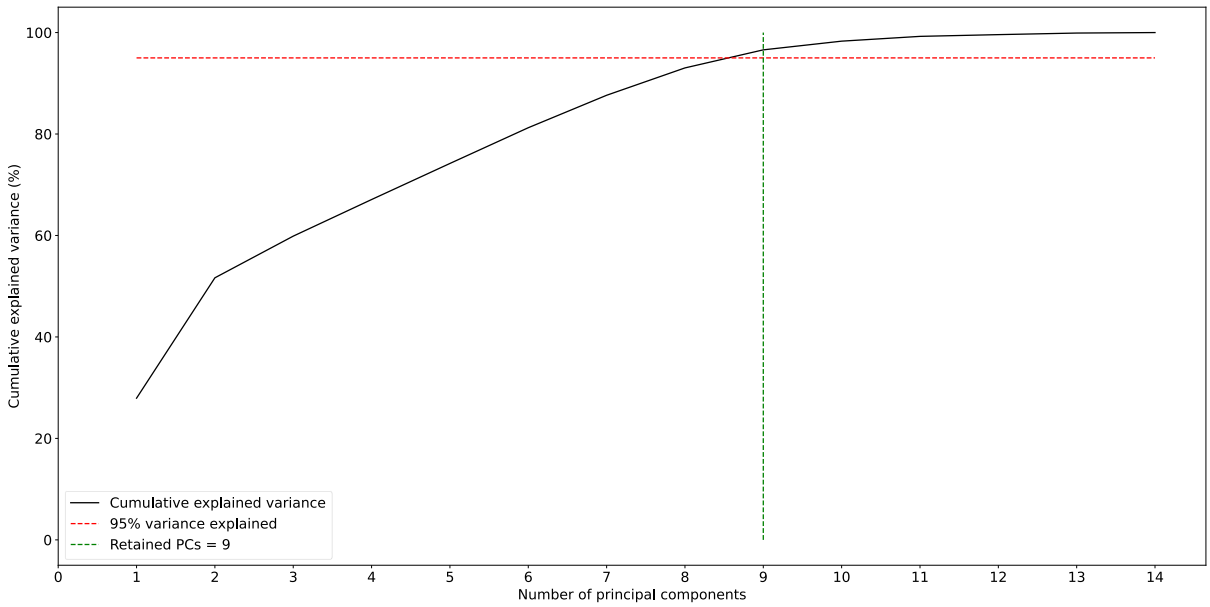


Figure 2. A plot of the cumulative explained variance: the sum of all eigenvalues up to the corresponding principal component over the total sum of all eigenvalues. PCA was performed on the results from simulating a CSTR at normal operating conditions with control as well as process and measurement noise.

A less common method employs a hard threshold for singular values, where all singular values smaller than the limit are associated with the PCA model's residual space. One method determines the optimal hard threshold (OHT) for non-square observation matrix with unknown noise as (Gavish & Donoho, 2014: 5042 Equation 4)

$$\hat{\tau}_* = \omega(\beta) S_{med}, \qquad \textbf{(11)}$$

using the median singular value S_{med} and the OHT coefficient $\omega(\beta)$, which is determined numerically and whose derivation will not be explored in this research. To calculate it, a MATLAB script provided by Gavish and Donoho is used (2014). An example of this approach is shown in Figure 3.

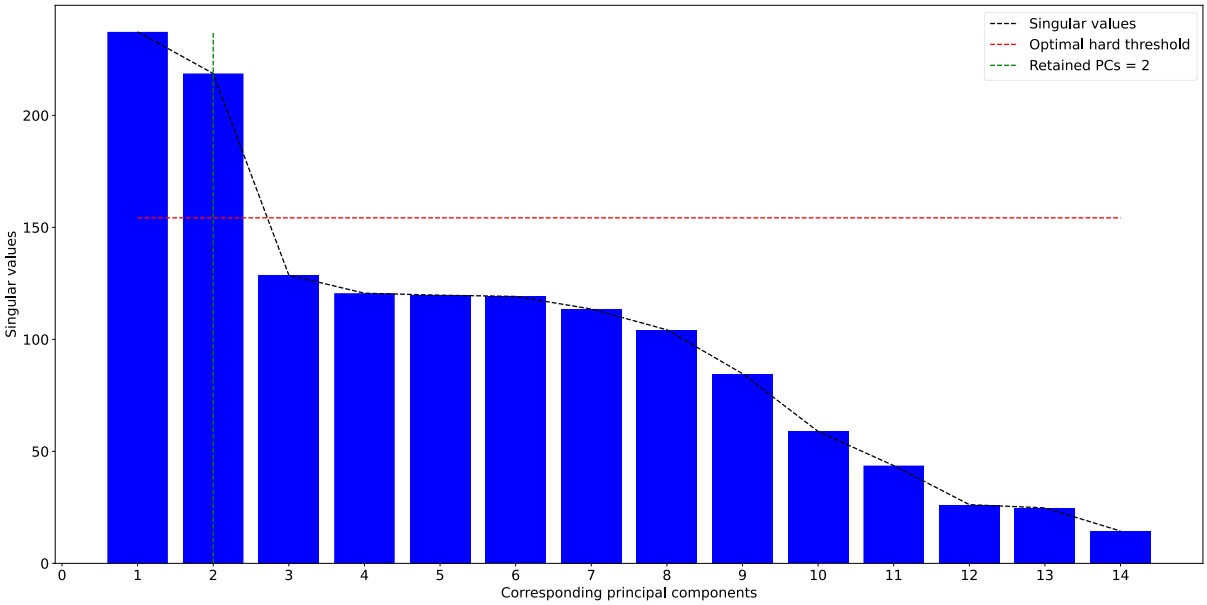


Figure 3. A plot of each singular value for each principal component in order from largest to smallest. PCA was performed on the results from simulating a CSTR at normal operating conditions with control as well as process and measurement noise.

3.1.3 Process monitoring and fault detection

3.1.3.1 Pattern-matching statistics

To detect abnormal data, Hotelling's T^2 statistic may be used. For each observation's scores ($\underline{\boldsymbol{t}}$) we have

$$T^2 = \underline{\boldsymbol{t}}^{\mathrm{T}} \underline{\boldsymbol{C}}^{-1} \underline{\boldsymbol{t}}, \tag{12}$$

with only the first k (subscript 'k') singular values used. Before assessing the T^2 statistic, the validity of the data should be checked using the squared prediction error (SPE), known also as the Q-statistic

$$SPE = \underline{\boldsymbol{e}}^{\mathrm{T}} \underline{\boldsymbol{e}}, \tag{13}$$

where $\underline{\mathbf{e}}$ is the row vector of prediction errors for an observation

$$\underline{\boldsymbol{e}} = \underline{\boldsymbol{x}} - \underline{\boldsymbol{P}}\underline{\boldsymbol{t}}. \tag{14}$$

The confidence limits of these statistics (subscript 'lim') at a chosen significance level 'lim' can be used to detect faults. Hotelling's T^2 statistic follows the F distribution with (Russell *et al.*, 2000: 20 Equation 2.11)

$$T^2_{\mathrm{lim}} = \frac{k(N-1)(N+1)}{N(N-k)} F_{\mathrm{lim}}(k, N-k). \tag{15}$$

The Q-statistic's confidence limit is found by (Jackson & Mudholkar, 1979: 342 Equation 3.4)

$$SPE_{\lim} = \frac{\theta_1}{h_0}\left(\frac{c_{\lim}\sqrt{2\theta_2 h_0^2}}{\theta_1} + 1 + \frac{\theta_2 h_0(h_0-1)}{h_0^2}\right), \quad \textbf{(16)}$$

where (for i = 1, 2, 3)

$$\theta_i = \sum_{j=k+1}^{M} S_j^i \quad \textbf{(17)}$$

and

$$h_0 = 1 - \frac{2\theta_1\theta_3}{3\theta_2^2}, \quad \textbf{(18)}$$

with c_{lim} being the one-tailed normal deviate calculated from the inverse cumulative normal distribution

$$c_{\lim} = \Phi^{-1}(1-\alpha). \quad \textbf{(19)}$$

For each observation, the contribution from each variable to the T^2 statistic can be found by

$$CONT_j = \sum_{i=1}^{D}\sum_{n=1}^{N} \underline{\boldsymbol{p}}_i^{\mathrm{T}} \underline{\boldsymbol{\lambda}}_k{}^{-1} \underline{\boldsymbol{t}}_n^{\mathrm{T}} x_{n,i}, \quad \textbf{(20)}$$

using $\underline{\boldsymbol{p}}_i$ as the ith column, and the diagonal matrix with the retained eigenvalues ($\underline{\boldsymbol{\lambda}}_k$). This provides the equivalent result to the equation provided by literature, where it should be noted that it is specified $p_{i,j}$ in literature, contrary to the expected convention of 'row, column' (Russell *et al.*, 2000; Rossouw, Coetzer & Le Roux, 2020). Where a fault is detected, the contribution that each variable makes to the T^2 statistic can be used to diagnose the source of the fault.

Additionally, Johannesmeyer et al. (2002) use a PCA similarity factor (Krzanowski, 1982)

$$S_{PCA} = \frac{trace\left(\underline{\boldsymbol{P}}_S^{\mathrm{T}} \underline{\boldsymbol{P}}_{\mathrm{H}} \underline{\boldsymbol{P}}_{\mathrm{H}}^{\mathrm{T}} \underline{\boldsymbol{P}}_{\mathrm{S}}\right)}{k} \quad \textbf{(21)}$$

which considers the principal subspaces of the snapshot (subscript 'S') and historical dataset (subscript 'H') respectively. They determined the best cut-off value for this statistic to be 0.965.

3.1.3.2 Fault detection

To detect faults in historical data, a snapshot of data corresponding to nominal data is required to train the PCA model. Thereafter, the historical data can be divided into arbitrarily (but equally) sized data to calculate Hotelling's T^2 statistic and the SPE with the PCA model from the nominal operation. The T^2 or Q statistic calculated for each measurement is compared to their respective confidence limit, and each time $T^2 > T^2_{lim}$ or SPE > SPE_{lim}, a violation is tallied for that statistic. The allowable number of violations r_{lim} is determined by using the inverse cumulative binomial distribution with a confidence

limit 'lim' and the number of measurements in the historical snapshot, with the single-trial success probability of the difference between unity and the confidence limit used for the statistic being tallied.

To evaluate the success of fault detection, either the number of data sets in the candidate pool that are actually similar (N_1) or not similar (N_2) must be known. The sum of N_1 and N_2 is N_P, the total number of datasets evaluated. The total number of data sets evaluated that are actually similar to the snapshot is N_{DB}. Statistics (Johannesmeyer *et al.*, 2002) like the pool accuracy

$$p \triangleq \frac{N_1}{N_p} \times 100\% \tag{22}$$

measures the model's ability to return true positives, and the pattern matching efficiency

$$\eta \triangleq \frac{N_1}{N_{DB}} \times 100\%, \tag{23}$$

measures the model's ability to detect every fault appearing in the dataset. It has a maximum of

$$\eta^{\max} \triangleq \begin{cases} \frac{N_P}{N_{DB}} \times 100\% & \text{for } N_p < N_{DB} \\ 100\% & \text{for } N_p \geq N_{DB} \end{cases} \tag{24}$$

These can be combined into the overall effectiveness

$$\xi \triangleq \frac{p + \eta}{2}. \tag{25}$$

Prioritising pool accuracy means that the model has very few false positives at the cost of missing some fault cases. This is useful for cases where it is not as critical for every fault occurrence to be detected, and the engineer does not want to waste time sifting through a large amount of data. Prioritising pattern matching efficiency means ensuring every fault occurrence is shortlisted, at the cost of more false positives. This is relevant for highly important faults where it is critical that the engineer finds every occurrence and does not mind sifting through a larger amount of data as a result. However, the best performance is characterised by both a high pool accuracy and a high pattern matching efficiency. This is the motivation for applying and judging performance with the overall effectiveness.

3.1.4 Applications to FDI

Multivariate statistical process control (MSPC) based on feature-extraction has seen considerable research efforts. Venkatasubramanian, Rengaswamy, Kavuri and Yin (2003) discuss fault diagnosis based on process history, split between qualitative and quantitative methods. Expert systems are one of the qualitative methods, and function based on codifying an expert's knowledge on a process's operation into a structure such as a decision tree (Rich, Venkatasubramanian, Nasrallah & Matteo, 1989). The other qualitative method discussed is qualitative trend analysis, which is a pattern recognition problem (Whiteley & Davis, 1992) that uses the trends in sensor signals when a past fault

occurred to detect and identify new cases of the same fault. The first of the quantitative feature extraction methods are the statistical techniques PCA and partial least squares. They share a core functionality of dimensionality reduction, but the partial least squares technique is used to model the latent variables that explain the variation in both process variables and connected variables such as product quality (Geladi & Kowalski, 1986). The other quantitative method is neural networks, whose namesake provides the best overview appropriate for this research's scope.

Ding, Zhang, Ding, Yin, Naik, Deng & Gui (2010) apply PCA for fault detection and demonstrate that off-set faults are better detected by using the residual subspace, instead of the principal subspace, to compute pattern-matching statistics. Off-set faults are those that cause a static departure from the nominal value and does not change with size or time. Hotelling's T^2 statistic and the Q statistic are often used for fault detection alongside control charts (Mnassri, Adel, Ananou & Ouladsine, 2009).

Several augmentations and formulations of PCA have been devised towards the goal of improving the fault detection ability of PCA. Dynamic PCA attempts to deal the correlation of variables across time by performing PCA on an observation matrix of time-shifted duplicate vectors (Ku, Storer & Georgakis, 1995). Probabilistic PCA was discovered by attempting to reduce the computational requirements of PCA through the use of an expectation-maximisation algorithm and placing probability distributions over each of the model parameters and variables (Roweis, 1997; Tipping & Bishop, 1999a). Kernel PCA uses kernel functions (Boser, Guyon & Vapnik, 1992) to enable PCA in a feature space of arbitrarily large dimensionality, which is computationally infeasible for normal PCA which scales on the order of D^3 (Schölkopf, Smola & Müller, 1997). Sensitive PCA has been devised as a framework for PCA fault detection that chooses the retained components as the PCs found to be most sensitive to faults (Jiang, Yan & Zhao, 2013).

Villegas et al.'s (2010) research communicates the importance of the choice of retained PCs and that the Q statistic performed well to identify faults using data from a two-tank simulation validated with plant data. However, Mnassri et al. (2009) share research that indicates that the T^2 and Q statistic struggle to perform in real industrial settings, showing that the Q and T^2 statistic are more prone to misclassifying the faulty variable through their contributions. Jiang et al. (2013) use the Tennessee Eastman process to test PCA-based FDI, and whose results indicate Hotelling's T^2 was unable to detect faults that sensitive PCA could. Ge and Song (2013b) devise a scheme to construct sub-blocks within the PCA feature space to create distributed PCA models for plant-wide control. They report that the distributed PCA models are ablet to reflect local changes in plant operation and also provide improved process monitoring by combining the monitoring results of the distributed models.

de Carvalho Michalski & Martha de Souza (2022) emphasizes base assumptions for PCA are not true for some dynamic systems. These assumptions are of independent and identically distributed variables that are not correlated in time. A simple example for where these assumptions fail is in a flash drum where pressure, temperature, and compositions are measured. Thermodynamics dictates that the system is fixed by only one of either temperature and pressure, and the outlet compositions are directly dependent on the inlet composition as well as the state of the system. Most chemical engineering processes suffer from the same dependence and autocorrelation.

There is research into other fault detection and monitoring approaches that do not use PCA, which include Bayesian networks (Tidriri, Chatti, Verron & Tiplica, 2016), autoencoders (Qian, Song, Yao, Zhu & Zhang, 2022), and wavelet analysis (Chen & Chen, 2025). Lu et al. (2003) use wavelet analysis to improve PCA fault detection as they find some fault types are more distinct in the frequency domain than the time domain.

3.2 Bayesian Principal Component Analysis

3.2.1 Model development

Due to the fact that the majority of existing research uses the convention that matrices consist of row vectors representing distinct variables, this convention will be used for matrices defined within this section. Similarly, the variables in this section largely do not overlap those of the previous section. Therefore, the appearance of previously defined symbols carries the assumption that they do, in fact, represent the same quantity, but with the convention adopted for this section.

Tipping and Bishop (1999a) and Roweis (1997) independently discovered probabilistic PCA (PPCA). Bishop then went on to extend his discovery to formulate BPCA (1998). Thus, most of the knowledge in this section comes from Bishop's Pattern Recognition and Machine Learning textbook (2006), wherein the concepts are thoroughly explored. However, the monitoring statistics are explained by Ge and Song's Multivariate Statistical Process Control textbook (Ge & Song, 2013a).

As one of the cornerstones of BPCA, one must first present Bayes' theorem given by

$$\mathrm{P}(x|y) = \frac{\mathrm{P}(y|x)\mathrm{P}(x)}{\mathrm{P}(y)}, \tag{26}$$

with arbitrary symbols x and y to represent a cause and effect, respectively. Bayes' theorem describes the how posterior evidence P(y) is used to update our prior beliefs P(x), dependent on a likelihood P(x|y). The likelihood evaluates how much sense the result makes based on our existing knowledge of the cause and effect.

As the basis of BPCA, PPCA models the relationship between the M-dimensional latent variable $\underline{\boldsymbol{z}}$ and D-dimensional observed variable $\underline{\boldsymbol{x}}$ (Bishop, 2006: 572 Equation 12.33) as

$$\underline{\boldsymbol{x}} = \underline{\boldsymbol{W}}\underline{\boldsymbol{z}} + \underline{\boldsymbol{\mu}} + \underline{\boldsymbol{\epsilon}}, \tag{27}$$

consisting of the matrix of model weights $\underline{\boldsymbol{W}} \in \mathbb{R}^{\mathrm{D}\times\mathrm{M}}$ and the mean observed values $\underline{\boldsymbol{\mu}}$. The Gaussian noise $\underline{\boldsymbol{\epsilon}}$ has zero-mean and isotropic covariance defined in terms of σ^2 and the identity matrix $\underline{\mathbf{I}}$ written as

$$\mathrm{P}(\underline{\boldsymbol{\epsilon}}) = \mathcal{N}(\underline{\boldsymbol{\epsilon}}|0, \sigma^2\underline{\boldsymbol{I}}). \tag{28}$$

The latent variables are also taken as Normally distributed with zero-mean, but with unit variance, as given by

$$\mathrm{P}(\underline{\boldsymbol{z}}) = \mathcal{N}(\underline{\boldsymbol{z}}|0, \underline{\boldsymbol{I}}). \tag{29}$$

Taking another step towards Bayes' theorem, the marginal probability of the observed variables is needed as a predictive distribution, which is given by

$$\mathrm{P}(\underline{\boldsymbol{x}}) = \mathcal{N}(\underline{\boldsymbol{x}}\,|\underline{\boldsymbol{\mu}}, \underline{\boldsymbol{K}}), \tag{30}$$

where the Gaussian distribution is defined as

$$\mathcal{N}(\underline{\boldsymbol{x}}\,|\underline{\boldsymbol{\mu}}, \underline{\boldsymbol{K}}) = \left(\frac{1}{2\pi}\right)^{\frac{D}{2}} \left(\frac{1}{|\underline{\boldsymbol{K}}|}\right)^{\frac{1}{2}} \exp\left(-\frac{1}{2}(\underline{\boldsymbol{x}} - \underline{\boldsymbol{\mu}})^{\mathrm{T}} \underline{\boldsymbol{K}}^{-1} (\underline{\boldsymbol{x}} - \underline{\boldsymbol{\mu}})\right). \tag{31}$$

This uses the covariance matrix defined as

$$\underline{\boldsymbol{K}} = \underline{\boldsymbol{W}}\underline{\boldsymbol{W}}^{\mathrm{T}} + \sigma^2 \underline{\boldsymbol{I}} \in \mathbb{R}^{\mathrm{D}\times\mathrm{D}}, \tag{32}$$

whose inverse can be calculated by (Bishop, 2006: 573 Equation 12.40)

$$\underline{\boldsymbol{K}}^{-1} = \sigma^{-1}\underline{\boldsymbol{I}} - \sigma^{-2}\underline{\boldsymbol{W}}\underline{\boldsymbol{M}}^{-1}\underline{\boldsymbol{W}}^{\mathrm{T}} \in \mathbb{R}^{\mathrm{D}\times\mathrm{D}}, \tag{33}$$

where we have (Bishop, 2006: 573 Equation 12.41)

$$\underline{\boldsymbol{M}} = \underline{\boldsymbol{W}}^{\mathrm{T}}\underline{\boldsymbol{W}} + \sigma^2 \underline{\boldsymbol{I}} \in \mathbb{R}^{\mathrm{M}\times\mathrm{M}}. \tag{34}$$

The distribution of the observed variables conditioned on the latent variables is

$$\mathrm{P}(\underline{\boldsymbol{x}}|\underline{\boldsymbol{z}}) = \mathcal{N}\left(\underline{\boldsymbol{x}}\middle|\underline{\boldsymbol{W}}\underline{\boldsymbol{z}} + \underline{\boldsymbol{\mu}}, \sigma^2 \underline{\boldsymbol{I}}\right). \tag{35}$$

Combining this with the marginal distribution from equation (29), the posterior distribution of the latent variables is also Gaussian and defined for linear-Gaussian models by (Bishop, 2006: 93, 573 Equation 2.116, Equation 12.42)

$$\begin{aligned}\mathrm{P}(\underline{\boldsymbol{z}}|\underline{\boldsymbol{x}}) &= \mathcal{N}\left(\underline{\boldsymbol{z}}|\left(\underline{\boldsymbol{I}} + \underline{\mathbf{W}}^{\mathrm{T}}\sigma^{-2}\underline{\boldsymbol{I}}\underline{\boldsymbol{W}}\right)^{-1}\left(\underline{\mathbf{W}}^{\mathrm{T}}\sigma^{-2}\underline{\boldsymbol{I}}(\underline{\boldsymbol{x}} - \underline{\boldsymbol{\mu}}) + \underline{\boldsymbol{I}}\underline{\boldsymbol{\mu}}\right),\right) \\ &= \mathcal{N}\left(\underline{\boldsymbol{z}}|\underline{\boldsymbol{M}}^{-1}\underline{\mathbf{W}}^{\mathrm{T}}(\underline{\boldsymbol{x}} - \underline{\boldsymbol{\mu}}), \sigma^{-2}\underline{\boldsymbol{M}}\right).\end{aligned} \tag{36}$$

The complete-data Bayes' theorem is given by

$$\mathrm{P}\left(\underline{\boldsymbol{\mu}}, \underline{\mathbf{W}}, \sigma^2\middle|\underline{\mathbf{X}}, \underline{\mathbf{Z}}\right) = \frac{\mathrm{P}\left(\underline{\mathbf{X}}, \underline{\mathbf{Z}}\middle|\underline{\boldsymbol{\mu}}, \underline{\mathbf{W}}, \sigma^2\right)\mathrm{P}\left(\underline{\boldsymbol{\mu}}, \underline{\mathbf{W}}, \sigma^2\right)}{\mathrm{P}(\underline{\mathbf{X}}, \underline{\mathbf{Z}})} \tag{37}$$

and treats the mean as a parameter to avoid placing a prior over it, but the exact maximum likelihood solution is already known to be the sample mean $\bar{\mathbf{x}}$ (Bishop, 2006: 578 Equation 12.52).

Because the number of data points is relatively large compared to the number of variables and the posterior distribution is tightly peaked the evidence approximation can be used (Tipping & Bishop,

1999b) to choose an independent zero-mean Gaussian prior over each column (subscript 'i') of $\underline{\mathbf{W}}$, each with their own hyperparameter α as the precision with

$$\mathrm{P}(\underline{\mathbf{w}}_i) = \mathcal{N}\left(\underline{\mathbf{w}}_i \middle| 0, \alpha_i^{-1}\right) \tag{38}$$

such that, using the Gaussian distribution's equation (31), we get

$$\mathrm{P}(\underline{\boldsymbol{W}}|\underline{\boldsymbol{\alpha}}) = \prod_{i=1}^{M} \mathcal{N}(\underline{\mathbf{w}}_i|\alpha_i) = \prod_{i=1}^{M} \left(\frac{\alpha_i}{2\pi}\right)^{\frac{D}{2}} \exp\left(-\frac{\alpha_i}{2}\underline{\boldsymbol{w}}_i^{\mathrm{T}}\underline{\boldsymbol{w}}_i\right). \tag{39}$$

The derivation of the expectation-maximisation (EM) algorithm used to learn the BPCA model parameters can bow begin. Assuming that data points are independent and using the likelihood function for a Gaussian (Bishop, 2006: 26 Equation 1.53), we have

$$\mathrm{P}\left(\underline{\mathbf{X}}, \underline{\mathbf{Z}} \middle| \underline{\boldsymbol{\mu}}, \underline{\mathbf{W}}, \sigma^2\right) = \prod_{n=1}^{N} \mathcal{N}\left(\underline{\boldsymbol{x}}, \underline{\boldsymbol{y}} \middle| \underline{\boldsymbol{\mu}}, \underline{\mathbf{W}}, \sigma^2\right) = \prod_{n=1}^{N} \mathrm{P}(\underline{\mathbf{x}}_{\mathrm{n}}|\underline{\mathbf{z}}_{\mathrm{n}})\mathrm{P}(\underline{\mathbf{z}}_{\mathrm{n}}) \tag{40}$$

has its log taken (Bishop, 2006: 578 Equation 12.52) to yield

$$\ln \mathrm{P}\left(\underline{\mathbf{X}}, \underline{\mathbf{Z}} \middle| \underline{\boldsymbol{\mu}}, \underline{\mathbf{W}}, \sigma^2\right) = \sum_{n=1}^{N} \{\ln \mathrm{P}(\underline{\mathbf{x}}_{\mathrm{n}}|\underline{\mathbf{z}}_{\mathrm{n}}) + \ln \mathrm{P}(\underline{\mathbf{z}}_{\mathrm{n}})\} \tag{41}$$

as this creates a monotonically increasing function to maximise and improves the numerical stability of the optimisation by working with the sum of small numbers instead of their product.

For the expectation step of the EM algorithm, equation (41) has its expectation taken with respect to the posterior distribution over the latent variable. Substituting for the Gaussian distribution's equation (31) and simplifying produces

$$\begin{aligned} &\mathbb{E}\left[\ln p\left(\underline{\boldsymbol{X}}, \underline{\boldsymbol{Z}} \middle| \underline{\boldsymbol{\mu}}, \underline{\boldsymbol{W}}, \sigma^2\right)\right] \\ &\quad = -\sum_{n=1}^{N} \left\{\frac{D}{2}\ln(2\pi\sigma^2) + \frac{1}{2}\mathrm{Tr}\left(\mathbb{E}[\underline{\boldsymbol{z}}_n\underline{\boldsymbol{z}}_n^{\mathrm{T}}]\right) + \frac{1}{2\sigma^2}\left\|\underline{\boldsymbol{x}}_n - \underline{\boldsymbol{\mu}}\right\|^2 \right. \\ &\quad \left. - \frac{1}{\sigma^2}\mathbb{E}[\boldsymbol{z}_n]^T\underline{\boldsymbol{W}}^{\mathrm{T}}\left(\underline{\boldsymbol{x}}_n - \underline{\boldsymbol{\mu}}\right) + \frac{1}{2\sigma^2}\mathrm{Tr}\left(\mathbb{E}[\underline{\boldsymbol{z}}_n\underline{\boldsymbol{z}}_n^{\mathrm{T}}]\underline{\boldsymbol{W}}^{\mathrm{T}}\underline{\boldsymbol{W}}\right)\right\}. \end{aligned} \tag{42}$$

Only the sufficient statistics of the Gaussian distribution are required are required to calculate the expected value of the log likelihood, which are given by

$$\begin{aligned} &\mathbb{E}[\boldsymbol{z}_n] = \underline{\boldsymbol{M}}^{-1}\underline{\boldsymbol{W}}^{\mathrm{T}}(\underline{\boldsymbol{x}}_n - \overline{\underline{\boldsymbol{x}}}), \\ &\mathbb{E}[\underline{\boldsymbol{z}}_n\underline{\boldsymbol{z}}_n^{\mathrm{T}}] = \sigma^2\underline{\boldsymbol{M}}^{-1} + \mathbb{E}\left[\underline{\boldsymbol{z}}_n\right]\mathbb{E}[\underline{\boldsymbol{z}}_n]^T. \end{aligned} \tag{43}$$

For the maximisation step of the EM algorithm, equation (42) is maximised by taking first taking the derivative with respect to $\underline{\boldsymbol{W}}$ (Bishop, 2006: 583 Equation 12.63), shown by

$$\underline{\boldsymbol{W}}_{new} = \left[\sum_{n=1}^{N}(\underline{\boldsymbol{x}}_n - \overline{\underline{\boldsymbol{x}}})\mathbb{E}[\underline{\boldsymbol{z}}_n]^T\right]\left[\sum_{n=1}^{N}\mathbb{E}[\underline{\boldsymbol{z}}_n\underline{\boldsymbol{z}}_n^T] + \sigma^2\mathrm{diag}(\underline{\boldsymbol{\alpha}})\right]^{-1}, \tag{44}$$

which uses the property given as (Bishop, 2006: 698 Equation C.24)

$$\frac{\partial}{\partial \underline{\boldsymbol{A}}}\mathrm{Tr}(\underline{\boldsymbol{A}}\underline{\boldsymbol{B}}) = \underline{\boldsymbol{B}}. \tag{45}$$

Then, the derivative taken with respect to σ^2 yields

$$\sigma^2_{new} = \frac{1}{ND}\sum_{n=1}^{N}\{(\underline{\boldsymbol{x}}_n - \overline{\underline{\boldsymbol{x}}})(\underline{\boldsymbol{x}}_n - \overline{\underline{\boldsymbol{x}}})^{\mathrm{T}} - 2\mathbb{E}[\underline{\boldsymbol{z}}_n]^{\mathrm{T}}\underline{\boldsymbol{W}}_{\mathrm{new}}^{\mathrm{T}}(\underline{\boldsymbol{x}}_n - \overline{\underline{\boldsymbol{x}}}) + \mathrm{Tr}\big(\mathbb{E}[\underline{\boldsymbol{z}}_n\underline{\boldsymbol{z}}_n^{\mathrm{T}}]\underline{\boldsymbol{W}}_{\mathrm{new}}^{\mathrm{T}}\underline{\boldsymbol{W}}_{\mathrm{new}}\big)\}. \tag{46}$$

To determine the new values for the hyperparameters, the log marginal likelihood with $\underline{\mathbf{W}}$ is integrated out (Bishop, 2006: 582 Equation 12.61) to get

$$\ln \mathrm{P}\big(\underline{\mathbf{X}}\big|\underline{\boldsymbol{\alpha}}, \underline{\boldsymbol{\mu}}, \sigma^2\big) = \ln \int \mathrm{P}\big(\underline{\mathbf{X}}\big|\underline{\boldsymbol{W}}, \underline{\boldsymbol{\mu}}, \sigma^2\big)\mathrm{P}(\underline{\boldsymbol{W}}|\underline{\boldsymbol{\alpha}})d\underline{\mathbf{W}}, \tag{47}$$

which is maximised, where the probability of the data conditioned on the parameters is (Bishop, 2006: 574 Equation 12.43)

$$\mathrm{P}\big(\underline{\mathbf{X}}\big|\underline{\boldsymbol{W}}, \underline{\boldsymbol{\mu}}, \sigma^2\big) = \sum_{n=1}^{N}\ln \mathrm{P}\big(\underline{\boldsymbol{x}}_{\mathrm{n}}\big|\underline{\boldsymbol{W}}, \underline{\boldsymbol{\mu}}, \sigma^2\big) = -\frac{ND}{2}\ln 2\pi - \frac{N}{2}\ln|\underline{\boldsymbol{K}}| - \frac{1}{2}\sum_{n=1}^{N}\big(\underline{\boldsymbol{x}}_n - \underline{\boldsymbol{\mu}}\big)^{\mathrm{T}}\underline{\boldsymbol{K}}^{-1}\big(\underline{\boldsymbol{x}}_n - \underline{\boldsymbol{\mu}}\big). \tag{48}$$

We must then assume a sharply peaked posterior distribution where the mode accurately captures the bulk of the probability mass and the mode is not skewed, allowing the Laplace approximation to be used. This approximation is required due to equation (47) being intractable, and results in

$$\alpha_i^{\mathrm{new}} = \frac{D}{\underline{\boldsymbol{w}}_i^{\mathrm{T}}\underline{\boldsymbol{w}}_i}. \tag{49}$$

3.2.2 Process monitoring and fault detection

Adjusting for BPCA, Hotelling's T^2 can now be defined by (Ge & Song, 2013a: 151 Equation 11.20)

$$T^2 = \underline{\boldsymbol{t}}^{\mathrm{T}}\mathrm{var}^{-1}(\underline{\boldsymbol{t}}|\underline{\boldsymbol{x}})\underline{\boldsymbol{t}}, \tag{50}$$

with the score vectors

$$\underline{\boldsymbol{t}} = \underline{\boldsymbol{P}}^{\mathrm{T}}\left(\underline{\boldsymbol{P}}\underline{\boldsymbol{P}}^{\mathrm{T}} + \sigma^2\underline{\boldsymbol{I}}\right)^{-1}\underline{\boldsymbol{x}} \quad \textbf{(51)}$$

and variance

$$\mathrm{var}(\underline{\boldsymbol{t}}|\underline{\boldsymbol{x}}) = \sigma^2\underline{\boldsymbol{M}}, \quad \textbf{(52)}$$

with the matrix of k retained principle components $\underline{\boldsymbol{P}} \in \mathbb{R}^{\mathrm{D}\times\mathrm{k}}$ defined the same as before. The Q statistic's definition remains unaffected by the new probabilistic formulation, other than the new score vector definition. The confidence limit for Hotelling's T^2 are now found using the chi-squared distribution at a chosen significance 'lim'

$$T^2_{\mathrm{lim}} = \mathcal{X}^2_{\mathrm{lim}}(k). \quad \textbf{(53)}$$

Similarly, the confidence limit for Q is

$$SPE_{\mathrm{lim}} = \mathrm{mean}(SPE_{\mathrm{S}})\mathcal{X}^2_{\mathrm{lim}}\left(\mathrm{var}(SPE_{\mathrm{S}})\right), \quad \textbf{(54)}$$

where the mean and variance of the Q statistic for the normal operating condition's BPCA model is used.

The contributions from each variable to Hotelling's T^2 for BPCA is based on reforming equation (20) to meet the conventions of this section

$$CONT_j = \sum_{i=1}^{D}\sum_{n=1}^{N}\underline{\boldsymbol{t}}_i^{part,\mathrm{T}}\mathrm{var}^{-1}(\underline{\boldsymbol{t}}|\underline{\boldsymbol{x}})\underline{\boldsymbol{t}}_n x_{n,i}, \quad \textbf{(55)}$$

where

$$\underline{\boldsymbol{t}}^{part,\mathrm{T}} = \left(\underline{\boldsymbol{P}}^{\mathrm{T}}\left(\underline{\boldsymbol{P}}\underline{\boldsymbol{P}}^{\mathrm{T}} + \sigma^2\underline{\boldsymbol{I}}\right)^{-1}\right)^{\mathrm{T}}. \quad \textbf{(56)}$$

3.2.3 Applications to FDI

Data reconciliation is important (Venkatasubramanian *et al.*, 2003), and BPCA allows the for reconstruction of missing points (Bishop, 1998). Additionally, BPCA allows for automatic dimensionality selection, which helps to resolve the model complexity aspect of PCA. Bouveyron et al. (2020) encourage the consideration of several dimensionality selection methods (while acknowledging the need for guidance to choose PCs), as their results show that both their method and several alternatives perform well across a wide range of signal-to-noise ratios. However, it would also be important to note the collapse of performance of all methods with a low signal-to-noise ratio.

There is a significant amount of research surrounding the application of Bayesian formulations of PCA to analyse genetic data in the biomedical field. Sartini, Zhou, Selvin, Zeger and Crainiceanu (2026) present Fast Bayesian Functional PCA (FBFPCA) to analyse a continuous glucose monitoring dataset that is aggregated by participant and Multilevel Fast PCA to individual meal response curves. FBFPCA

was shown to be very stable and perform better than existing methods used to analyse the large and complicated dataset. Yuan and Mancuso (2023) introduce Sum of Single Effects PCA (SuSiE PCA) as a Bayesian formulation of sparse PCA, and apply it to "multi-tissue expression quantitative trait loci data". They find it performs 18 times faster than sparse PCA and identifies more modules with a higher enrichment of ribosome-related genes. Jiang, Elrod, Kim, Swafford, Knight and Thompson (2022) developed multivariate Sparse Functional PCA as a Bayesian algorithm to allow microbiome researchers to better "model the temporal dynamics of multiple complex, nonlinear outcome trajectories simultaneously." Their new algorithm allows for the simultaneous estimation of multiple trajectories and demonstrates its ability to "analyse the microbial ratio between Lachnospiraceae and Bacteroides", capturing the desired trends in an application important to the prediction accuracy of type 2 diabetes.

Gai, Li & Stevenson (2008) formulate Robust Bayesian PCA that uses the Student's T-distribution instead of a Gaussian distribution over the noise in an attempt to answer how to choose the retained PCs for Robust PCA. Their approach mirrors that of the formulation of BPCA, but with the EM algorithm and associated parameter estimation derived for the Student's T-distribution. They report achieving the desired model pruning, and an increase of both the correctness of number of PCs chosen and the accuracy of recovered PCs.

Nakajima et al. (2011) discuss advantages and disadvantages of the two formulations of BPCA. They define partial Bayesian treatment as is described in Bishop's textbook (2006) and used in this research, remarking that it is advantageous over variational BPCA (VBPCA) in practice due to computational efficiency and provide an analytic-form solution. Nakajima et al. (2011) go on to remark on the popularity of VBPCA in practice due to its generality, but that it suffers from the required iterations for a solution. Furthermore, they present a simplified variational BPCA formulation that addresses the original's drawbacks of computational expense to becomes the most practically advantageous.

The popularity of VBPCA is certainly felt in literature. Ge and Song (2010) develop a mode localisation approach for the application of mixture VBPCA to multimodal process monitoring. They use a propylene production process study to prove mixture VBPCA is able to detect faults that mixture PPCA is unable to, and that it can also correctly identify the process mode. Similarly, Cao, Jan, Huang, Fang, Wang and Gui (2021) apply VBPCA and Kullback-Leibler divergence between mixture models to multimodal process monitoring. They compare their proposed technique to a Gaussian mixture model (GMM), GMM PCA, multimodal PCA, and other techniques applied to the same cases. They first apply their proposed technique to a pair of simulated CSTRs and find it is both better at detecting faults and also has fewer false positives than the comparable techniques. Thereafter, they apply the proposed technique to an industrial hydrocracking process case study, finding their technique results in fewer false alarms while detecting faults as quickly as comparable techniques, which they attribute to a better model of the process's multimodal nature. Liu et al. (2014) consider the application for process monitoring and fault detection of a wastewater treatment plant (WWTP). They exploit the strengths of VBPCA to fill in missing data to simulate rural WWTPs which are not well equipped with sensors, leading to VBPCA outperforming PCA in pattern matching using the T^2 and Q statistics.

4 Methodology

Due to the large number of symbols introduced in a short space, not all of the symbols used in this section are defined. Please refer to the nomenclature table in this report's preamble for the definitions of any undefined symbols. All code used for this project is made available in a public GitHub repository accessible via the permalink https://github.com/mivobd/Scripsy, also included in Appendix D.

4.1 CSTR Model

Johannesmeyer (1999) devised the model and control parameters for a jacketed CSTR process, shown by Figure 4. This was then used by Johannesmeyer et al. (2002) to devise a data generation strategy to test a novel PCA pattern matching technique. Unfortunately, the author was unable to source model values from Singhal's PhD thesis (2002), as it calls for reference to Johannesmeyer's Master's dissertation (1999) to acquire the needed parameters, but the Stellenbosch University library was unable to provide access this document. However, model values were used from Wong's Master's dissertation (2006), who used the same CSTR model from Johannesmeyer et al. (2002: 2028 Figure 2).

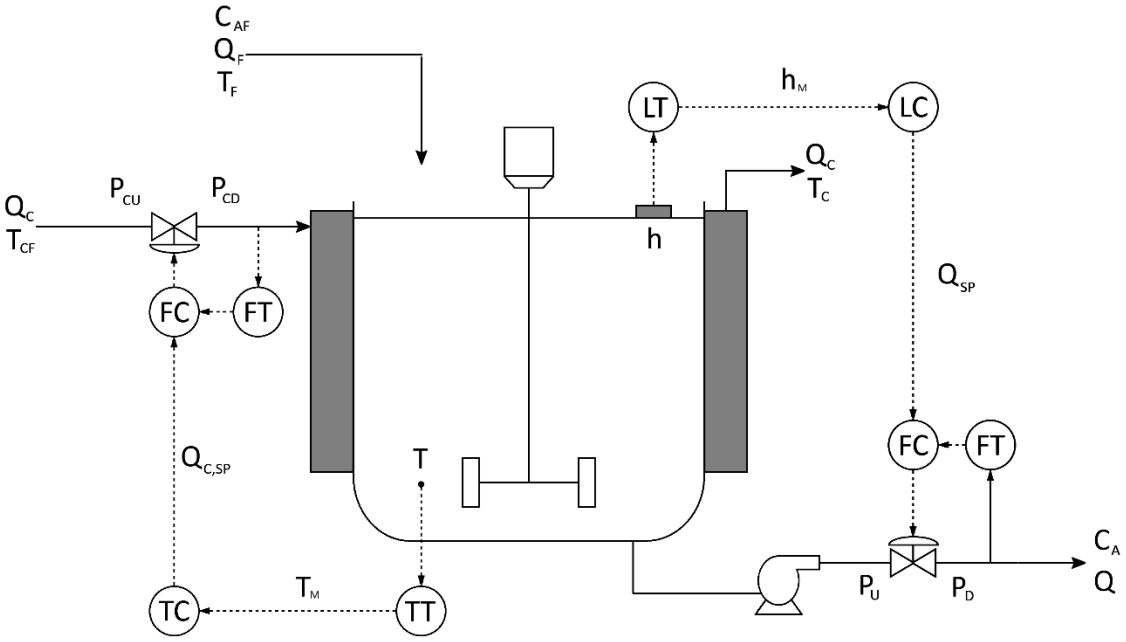


Figure 4. A diagram of the jacketed CSTR, redrawn from Johannesmeyer et al. (2002).

The model was developed using Simulink™ in the MATLAB R2026a release by representing the four conservation equations from Johannesmeyer et al. (2002: 2028)

$$\frac{dC_A}{dt} = -k_0 \exp\left(-\frac{E}{RT}\right) C_A + \frac{Q_F C_{AF} - Q C_A}{Ah} \qquad \textbf{(57)}$$

$$\frac{dT}{dt} = \frac{k_0 \exp\left(-\frac{E}{RT}\right) C_A(-\Delta H)}{\rho C_p} + \frac{Q_F T_F - QT}{Ah} + \frac{UA_C(T_C - T)}{\rho C_p Ah} \quad \textbf{(58)}$$

$$\frac{dT_C}{dt} = \frac{Q_C(T_{CF} - T_C)}{V_C} + \frac{UA_C(T - T_C)}{\rho C_{pC} Ah} \quad \textbf{(59)}$$

$$\frac{dh}{dt} = \frac{Q_F - Q}{A} \quad \textbf{(60)}$$

using Simulink™ blocks. Additionally, the valves for the reactor outlet (Q) and coolant (Q_C) flowrate were described by

$$Q = C_{vQ} l_Q N_{CF} \sqrt{\frac{P_U - P_D}{g_s}} \quad \textbf{(61)}$$

and

$$Q_C = C_{vQC} l_{QC} N_{CF} \sqrt{\frac{P_{CU} - P_{CD}}{g_s}}, \quad \textbf{(62)}$$

respectively. The specific gravity (g_s) was not specified and has been assumed to be unity (1) thus far, do to most liquids' density being close to that of water and due to the minor effect of changing this value on the overall aim to evaluate pattern matching. The valve level (l) appears explicitly in equation (61) and (62) due to the assumption of a linear valve characteristic (Seborg, Thomas F. Edgar, Duncan A. Mellichamp, & Francis J. Doyle III, 2016)

$$f(l) = l. \quad \textbf{(63)}$$

and the pressure drop is assumed to remain constant over the entire range of the valve. Due to questionable results for the valve level (resulting in an unstable process) using the values provided by Wong (2006), the valves were resized. This was done by finding the valve coefficient (C_v) for each valve at the nominal flow rate and assuming a steady state (subscript 'SS') valve level of 0.5 (half open) (Seborg *et al.*, 2016)

$$C_v = \frac{Q_{SS}}{0.5 N_{CF} \sqrt{\frac{\Delta P_{SS}}{g_s}}}, \quad \textbf{(64)}$$

where N_{CF} is a unit conversion factor of 0.0456 L min^{-1} $Pa^{-0.5}$. The dynamics for both of the valves (subscript 'v') were modelled by the transfer function (G)

$$G_v = \frac{K_v}{\tau_v s + 1} \quad \textbf{(65)}$$

with a gain (K) and time constant (τ) of 0.0625 mA. The nominal process values are provided under Table 1. To simulate realistic process operation and measurements, autoregressive and standard Gaussian noise was applied to the variables listed under Table 2. All variables whose value was saved had measurement noise applied, which was applied prior to control calculations where applicable.

Table 1. Nominal values for process variables from Johannesmeyer et al. (2002).

Variable	Nominal value	Variable	Nominal value
Q	100 L min^{-1}	A	16.66 dm^2
Q_C	15 L min^{-1}	k_0	7.2×10^{10} min^{-1}
T_F	320 K	ΔH	-5×10^4 J mol^{-1}
T_{CF}	300 K	ρC_p	239 J L^{-1} K^{-1}
T	402.35 K	$\rho_C C_{pC}$	4175 J L^{-1} K^{-1}
T_C	345.44 K	E/R	8750 K
C_{AF}	1.0 mol L^{-1}	U_{AC}	5×10^4 J min^{-1} K^{-1}
C_A	0.037 mol L^{-1}	V_C	10 L
h	6 dm		

Table 2. The types and magnitude of noise variance applied to variables. Measurement noise variances are from Johannesmeyer et al. (2002) and process noise variances are from Wong (2006).

Variable	Autoregressive process noise	Gaussian measurement noise
P_D	398000^2 $(Pa)^2$	
P_{CU}	335000^2 $(Pa)^2$	
Q_F	0.817^2 $(L\ min^{-1})^2$	0.444 $(L\ min^{-1})^2$
T_F	0.817^2 $(K)^2$	0.444 $(K)^2$
C_{AF}	0.00577^2 $(mol\ L^{-1})^2$	5.56×10^{-6} $(mol\ L^{-1})^2$
T_{CF}	0.817^2 $(K)^2$	0.444 $(K)^2$
C_A		5.56×10^{-6} $(mol\ L^{-1})^2$
h		0.0160 $(dm)^2$
Q		0.444 $(L\ min^{-1})^2$
T		0.444 $(K)^2$
T_C		0.444 $(K)^2$
Q_C		0.0100 $(L\ min^{-1})^2$

As per Figure 4, cascade control was implemented. One cascade loop controlled the CSTR level in the primary loop, which sent the setpoint to the secondary controller that measured the outlet tank volumetric flow rate and controlled the position on the tank's outlet valve. The other cascade controller had the primary loop controlling the temperature in the CSTR, which provided the setpoint for the secondary controller measuring the inlet coolant volumetric flowrate and controlling the position of the inlet coolant flow line valve.

Controller tuning parameters were provided by Wong (2006: 28 Table 3.4) for PI controllers, shown in Table 3. However, the form of the controller equation was not explicitly shared by either Johannesmeyer et al. (2002) or Wong (2006). The equation's form was determined by simply testing the original controller values from Wong (2006: 28 Table 3.4) in both the parallel and ideal forms, and choosing the form that resulted in the most sensible control. This resulted in the ideal form of PI controllers being selected. However, the 'better' control was still unstable. Therefore, the controller parameters were retuned.

The controllers were retuned by disconnecting the cascade loop not being retuned, and by tuning the inner controller first before moving to the outer controller. Setpoint step changes were used to tune controller response. Each controller was tuned by first reducing the gain and increasing the integral time by a significant margin. Then, the gain was increased until the response started to become oscillatory, before the gain was decreased again to ensure stable control. Thereafter, the integral time was decreased until the response became oscillatory, before increasing the integral time to ensure stable responses. The new controller parameters are made available in

Table 4.

A discrete controller of the ideal form

$$K_c\left(1 + T_I \cdot t_s \frac{1}{z-1}\right) \tag{66}$$

was used with a sampling time (t_s) of 5 seconds. A noisy signal degrades control performance. To counteract this, a filter can be applied to the signal prior to the controller. Measurements are saved for statistical analysis prior to the application of a filter. A simple low-pass filter (subscript 'f') with transfer function (Seborg *et al.*, 2016)

$$G_f = \frac{1}{\tau_f s + 1} \tag{67}$$

was added before each controller, where the best value of τ_f is estimated as a multiple of the sample time by evaluating the control performance at different values. It is important to note that for any step where control performance is evaluated across different program executions, that the Gaussian noise blocks must have their repeatable results enabled via fixing their seed (and ensuring this is disabled before generating any data for pattern recognition purposes).

Table 3. The original controller parameters. Recreated from Wong (2006: 28 Table 3.4).

Parameter	**h-Q Cascade Controller**		**T-Q_C Cascade Controller**	
	Inner (Q) Loop	**Outer (h) Loop**	**Inner (Q_C) Loop**	**Outer (T) Loop**
K_C	0.2 mA min L^{-1}	–3.0 L min^{-1} dm^{-1}	0.2 mA min L^{-1}	–2.0 L min^{-1} K^{-1}
T_I (min)	1.7/60	1.5	1.7/60	52.2/60

Table 4. The retuned controller parameters.

Parameter	h-Q Cascade Controller		T-Q_C Cascade Controller	
	Inner (Q) Loop	**Outer (h) Loop**	**Inner (Q_C) Loop**	**Outer (T) Loop**
K_C	0.2/2.5 mA min L^{-1}	–3.0×15 L min^{-1} dm^{-1}	0.2 mA min L^{-1}	–2.0/2 L min^{-1} K^{-1}
T_I (min)	1.7/60×15	1.5/60×15	1.7/60×35	52.2/60

Following the description of data generated from Johannesmeyer et al. (2002), 120 minutes of operation were simulated each run. However, the first, instead of last, 34.75 minutes were always normal operation. Faults would be introduced at time t = 34.75 minutes, and the remaining 85.25 minutes were used as the data window for pattern matching. The entire simulated period was saved for diagnostic and visualisation purposes.

The list of faults and their descriptions is available under Table 5. Mirroring the protocol used by Johannesmeyer et al. (2002), a total of 463 runs were produced for each dataset; 54 were normal operation, and the rest had a random fault. The large number of runs is to reduce the risk of over-representing unlikely situations in the dataset. The program was designed such that important factors could be conveniently selected from the main MATLAB file, and the Simulink™ file would not need editing (except for enabling repeatable results for the random noise generation). The corresponding operation ID was stored with every run for evaluating pattern-matching performance. All simulation data were saved as .json files for easy access in the Python script which would perform pattern recognition.

The MATLAB script was adjusted to allow the generation of large new, but easily customizable, historical datasets. This was used for the production of the aforementioned 463 run data sets for both with and without control, as well as with and without process noise, for a total of four (4) different datasets. All datasets have measurement noise. However, the SVD algorithm fails to converge for not only the normal operating conditions (NOC) data without process noise and control, but also the NOC data with process noise but no control. Therefore, these cannot be used. Therefore, two (2) datasets remain to be used for the CSTR simulation case study.

Table 5. The list of faults, their identifier (IDs), description, and expected magnitude. Recreated from Johannesmeyer et al. (2002: 2029 Table 2)

ID	Operating Condition	Description	Nominal Value
N	Normal operation	Operation at the nominal conditions. No disturbances	N/A
F1	Catalyst deactivation	The activation energy ramps up	The ramp rate for E/R is +3 K min^{-1}
F2	Heat-exchanger fouling	The heat-transfer coefficient ramps down	The ramp rate for U_{AC} is – 125 J min^{-2} K^{-1}

F3	Dead coolant flow measurement	The coolant flow rate measurement stays at its last value	N/A

Table 5 continued…

ID	Operating Condition	Description	Nominal Value
F4, F_4	Bias in reactor temperature measurement	The reactor temperature measurement has a bias	± 4 K
F6, F_6	Step change in Q_F	Step change in the feed flow rate	10 L min^{-1}
F7, F_7	Ramp change in C_{AF}	The concentration ramps up or down	mol L^{-1} min^{-1}
F8, F_8	Ramp change in T_F	The feed temperature ramps up or down	The ramp rate is ± 0.1 K min^{-1}
F9, F_9	Ramp change in T_{CF}	The coolant feed temperature ramps up or down	The ramp rate is ± 0.1 K min^{-1}
F10, F_10	Step change in P_{CU}	Step change in the upstream pressure in the coolant line	± 17236.9 Pa
F11, F_11	Step change in P_D	Step change in the pressure in the reactor outlet line	± 34473.8 Pa
F12	Damped oscillations in feed flow rate	The feed flow rate changes as $\exp\left(-\frac{t}{33}\right)\sin\left(\frac{2\pi t}{10}\right)$ L min^{-1}	10 L min^{-1}
S1, S_1	Setpoint change in T	Setpoint change for the reactor temperature	± 3 K

4.2 PCA

Python 3.13.5 with the “uv” package manager was used to implement PCA as per the SVD definition of equation (8), using the “numpy” package’s “linalg.svd” function. As the main file grew, a testing took place in a separate, dedicated file in to protect the main codebase. Git source control was used to maintain a backup cloud copy of the project repository on Github and to track blame and progress through commits. Where possible, the implementation of algorithms was validated against public documentation (such as the T^2 statistic calculation on MATLAB’s PCA documentation page).

Multiple PCA models were made, each differing in the selection of retained PCs. The methods used to select the retained PCs are CEV (Figure 5), the OHT for singular values (Figure 6), and a Scree plot.

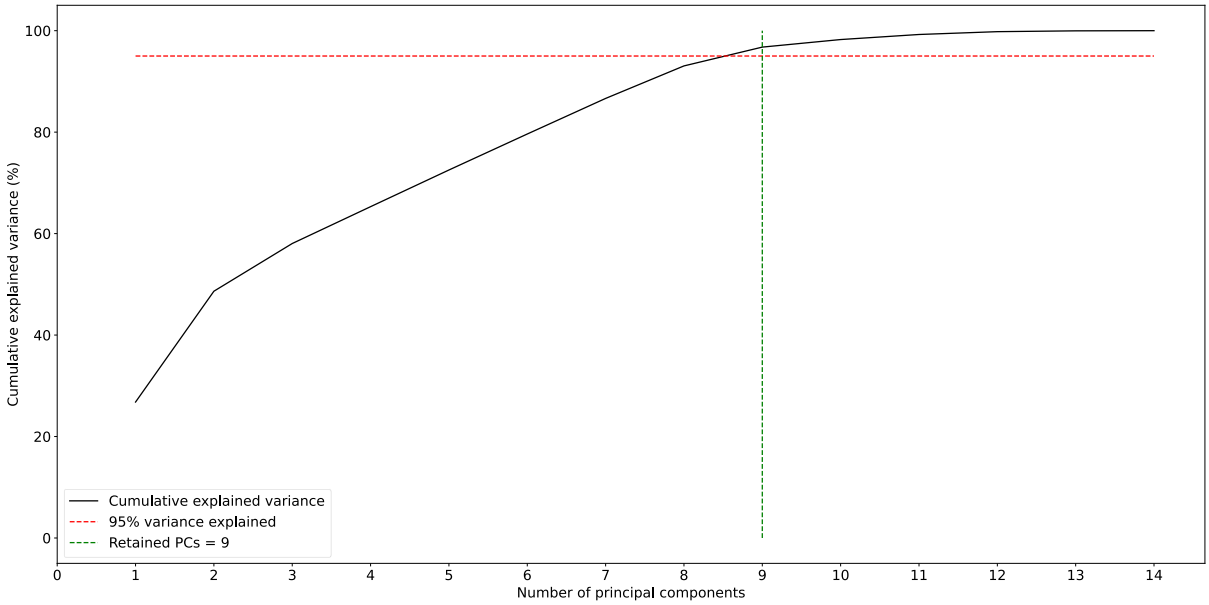


Figure 5. Selection of retained principal components for the principal component analysis model using a cumulative explained variance threshold. PCA was performed on the results from simulating a CSTR at normal operating conditions with measurement noise and control.

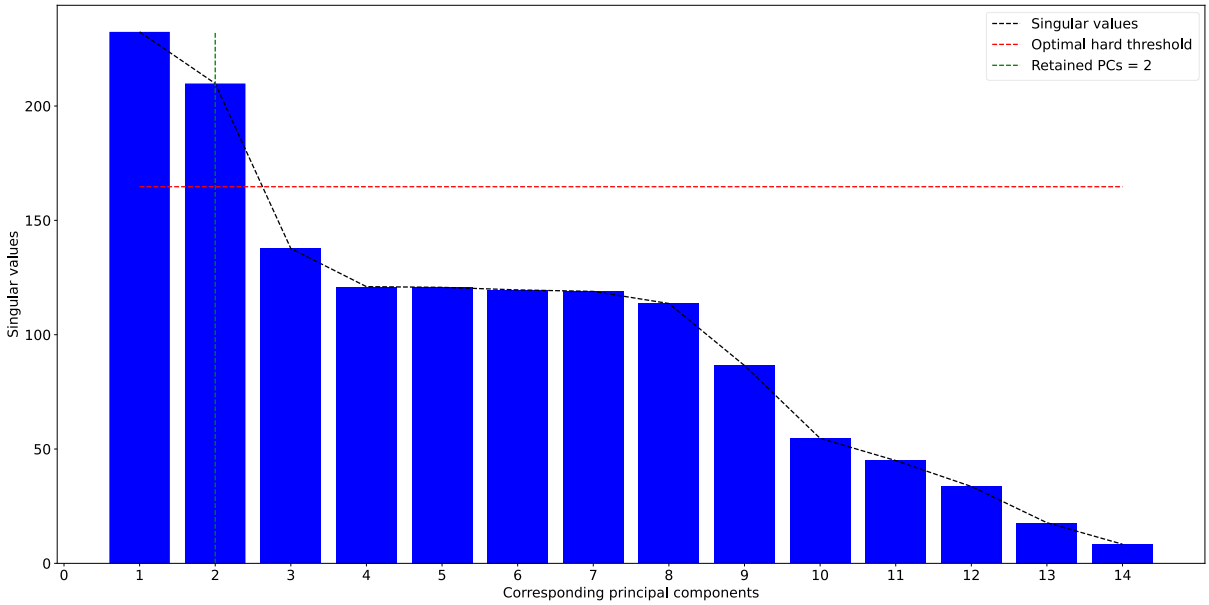


Figure 6. Choosing the retained principal components (PCs) to use in the principal component analysis model by discarding all PCs corresponding to a singular value below an optimal hard threshold. PCA was performed on the results from simulating a CSTR at normal operating conditions with measurement noise and control.

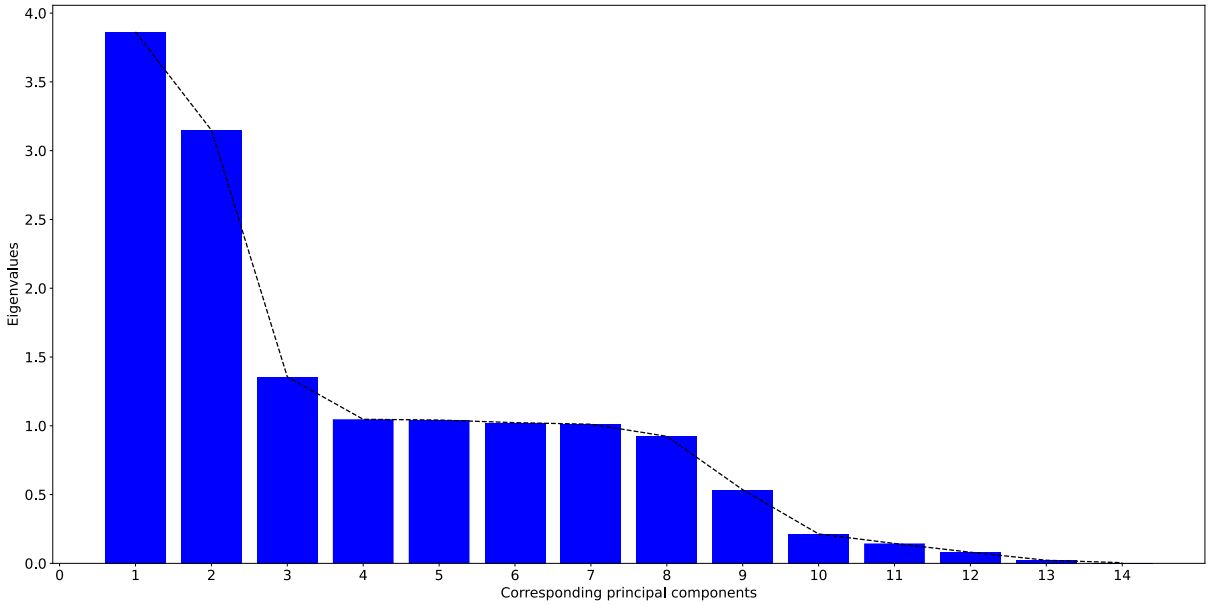


Figure 7. A Scree plot to determine the principal components to retain. PCA was performed on the results from simulating a CSTR at normal operating conditions with measurement noise and control.

To calculate the pattern matching statistics of equations (22), (24), and (25), each fault type was incrementally stored in a matrix as the script looped through every run's dataset. Using this record of fault IDs, the location (indices) of matching occurrences of the nominal 'N' fault ID can be found using Boolean logic, whose sum represents N_{DB}.

Similarly, while looping through each dataset, the number of violations of each statistic's confidence limit was stored. Using Boolean logic with both the violation limit r and the indices of actual fault occurrences found as described above, the indices of all correctly identified faults is acquired and its sum taken to get N_1. The efficiency metrics are then calculated as described in the theory.

4.3 BPCA

The equations of the EM algorithm were implemented in Python to learn the BPCA model for a set of simulation results. The EM algorithm was repeated 100 times with randomly generated seeds for each parameter, where the resulting model parameters that produced the largest log likelihood was retained as the best. Due to the time required for these calculations, a checkpoint system was implemented to save the progress of calculations. The random seeds for all runs are generated at the start and are saved along with the best parameters. The snapshot data also had a BPCA model trained across 1 000 initial guesses, which did not yield a better result than the first 100 guesses.

Due to no native pruning of principal components occurring, the amount of retained components was chosen using a method analogous to that of CEV. The analogy of explained variance in BPCA are the hyperparameters that explicitly rank the importance of their corresponding column in $\mathbf{W}$. The inverse value of hyperparameters are ordered from largest to smallest, as the inverse value of hyperparameters corresponds to whether a component is kept 'active' or not. Each of these values is compared to their total sum to create a cumulative distribution. Using a chosen cutoff value such as 95 %, the number of retained components can be found, as shown by Figure 8.

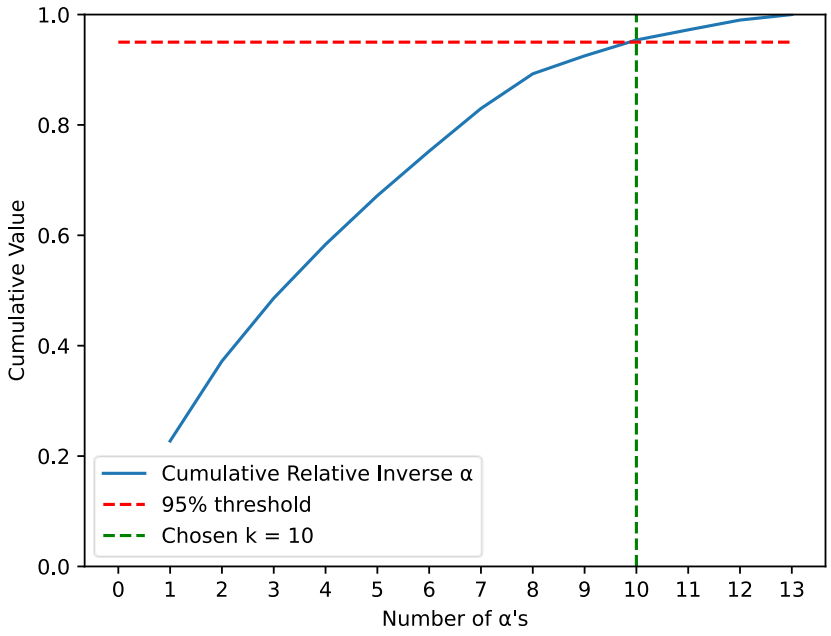


Figure 8. The cumulative relative inverse hyperparameter (α) distribution with a 95 % threshold to choose the number of retained principal components. Generated using the normal operation snapshot with active control and measurement noise, but without autoregressive process noise.

The calculation of pattern-matching efficiencies is carried using the same method employed for PCA.

5 Results and Discussion

5.1 Historical Data Generation

Bitmap vector images showing the developed Simulink model are available as Figure 25 and Figure 26 in the appendices. The model was validated by recreating figure A2.1 in Wong's dissertation (2006) by conducting the same step changes with no control or noise, shown by Figure 9.

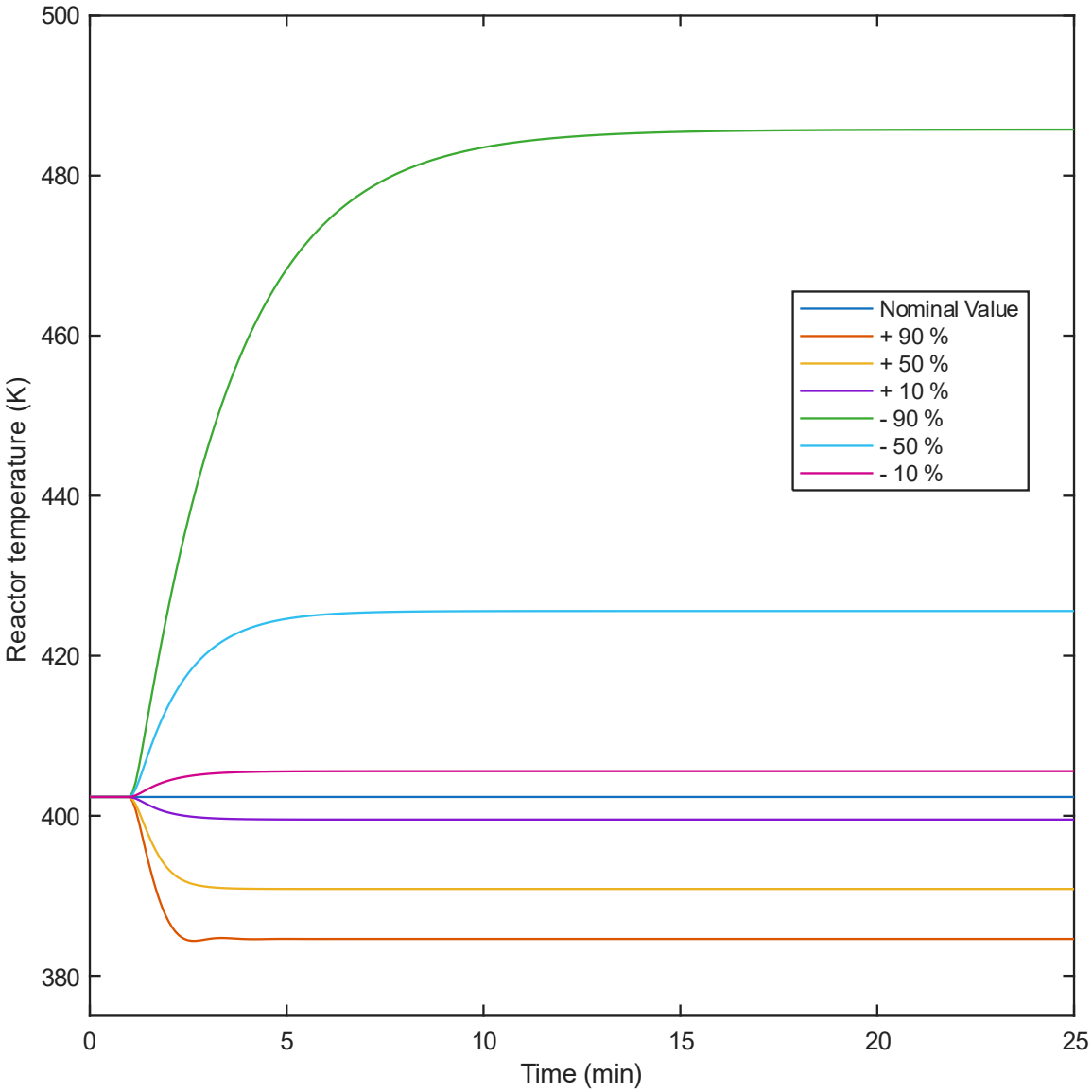


Figure 9. Results of the Simulink™ CSTR model validation. The effect on the reactor temperature by a step change to the coolant flow rate is tested. The step changes considered include ± 10 %, ± 50 %, and ± 90 % of the nominal value. Recreated from Wong (2006: 123 Figure A2.1) Appendix 2 Simulation Verification.

To better represent a real process, noise was added as both autoregressive input (upstream) process noise and scaled Gaussian measurement noise. However, like a real process, control action is degraded by noisy inputs. The signal to the controller has a filter applied to improve control action. Measurements are saved for statistical analysis prior to the application of a filter. The filter is described by equation (67), where the best value for τ_f was found to be 10 seconds, twice the sampling time. Values smaller caused increased noise in the measured signals, while larger values caused the process dynamics to become hidden.

The valve equation provided by Wong (2006) may have included a sign error in the exponent, as the simulation results which saw its use were not physically possible. It was decided that the best course of action would be to resize the valve, following Seborg et al.'s guidance (2016).

The controller parameters were retuned by heuristic fine-tuning; adjusting the gain so that the response was as fast as possible while still stable, before adjusting the integral time so that the rise time was decreased further, without causing a large overshoot response. The initial control performance is indicated with Figure 10, while the retuned performance is shown by Figure 11. To provide a reference to compare the control performance, Figure 12 shows the steady state operation of the CSTR with only measurement noise and no control. Using this reference state, the new level control performance is substantially improved while a large amount of variability remains in the reactor temperature due to the effect of reactor outlet flow rate on it.

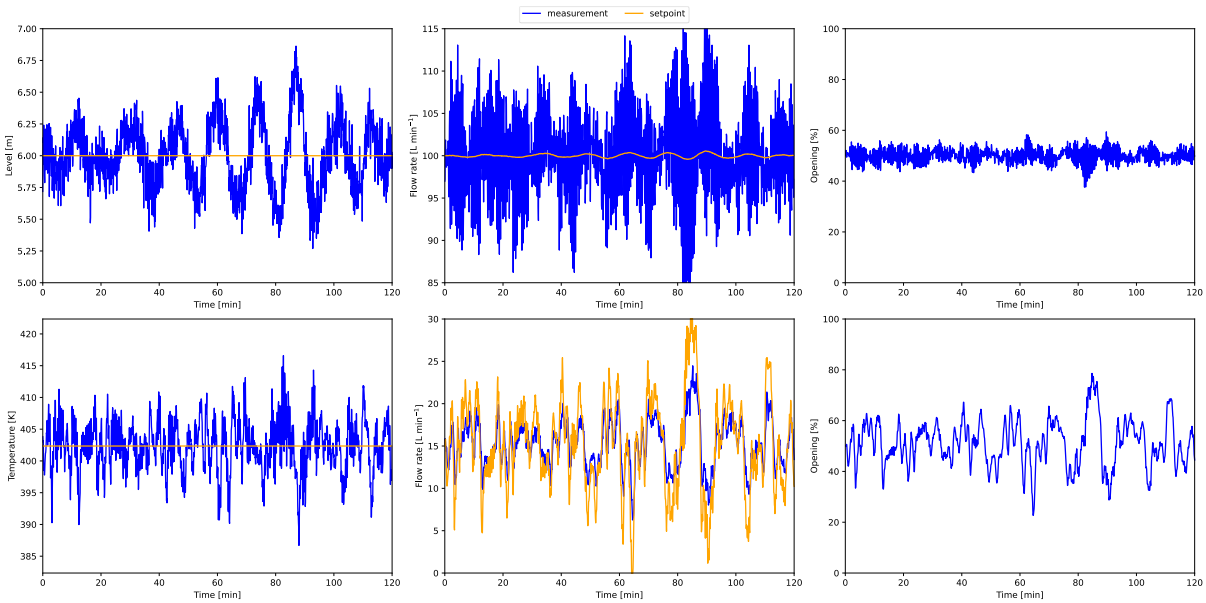


Figure 10. The control performance of the CSTR model using the tuning parameters from Wong (2006). The two rows of plots represent the two cascade control loops and their respective parameters as you progress from outer to inner loop from left to right; the primary controller's setpoint, the secondary controller's setpoint, and the final control element's position.

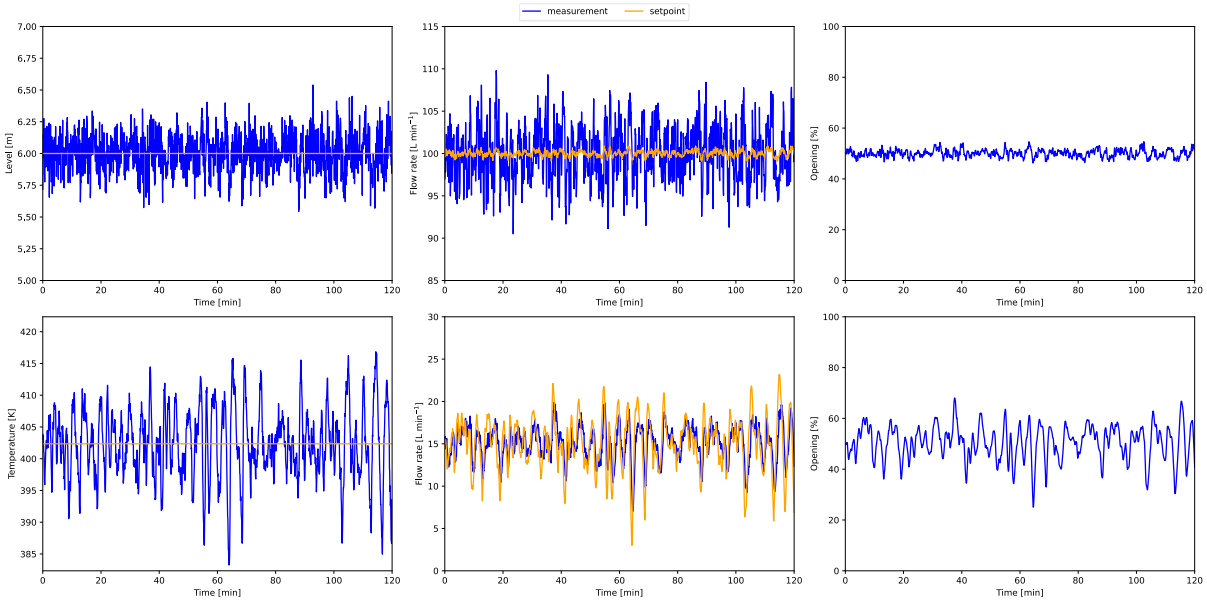


Figure 11. The control performance of the CSTR model after the controllers were retuned. The two rows of plots represent the two cascade control loops and their respective parameters as you progress from outer to inner loop from left to right; the primary controller's setpoint, the secondary controller's setpoint, and the final control element's position.

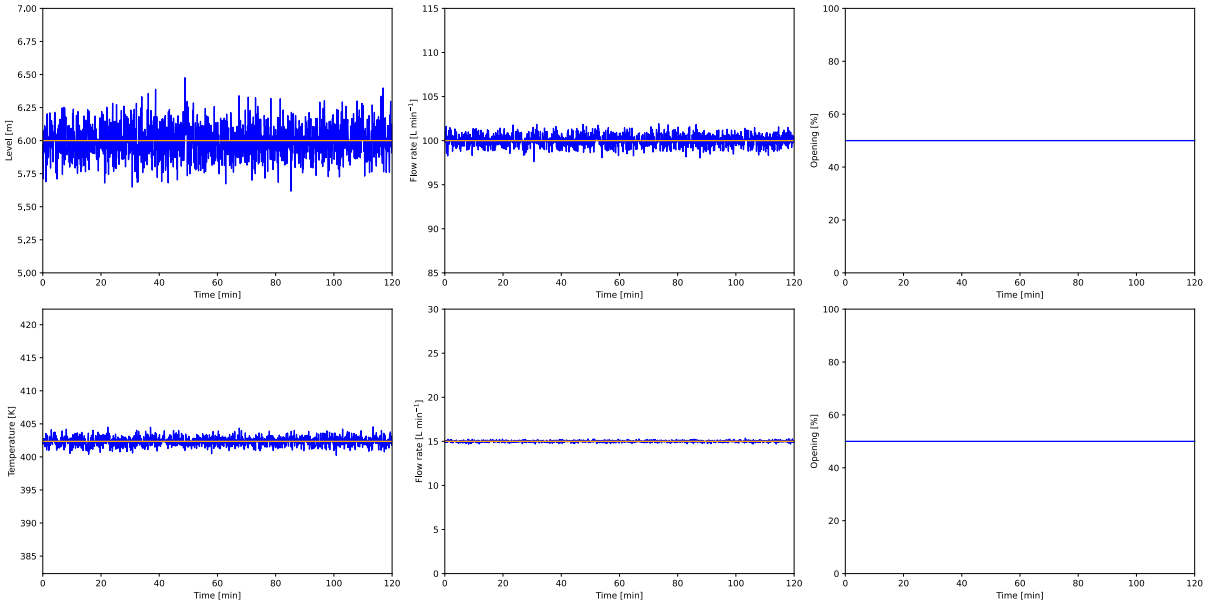


Figure 12. Steady state CSTR operation with measurement noise and no process noise or control to provide a reference for control performance. This data does not have the same noise seed as those in Figure 10 and Figure 11 as it was taken from a run used for a pattern-matching dataset. The two rows of plots represent the two cascade control loops and their respective parameters as you progress from outer to inner loop from left to right; the primary controller's setpoint, the secondary controller's setpoint, and the final control element's position.

5.2 PCA Fault Detection

5.2.1 Intermediate noise data

Pattern-matching was performed on the dataset without process noise, and with measurement noise and control (the 'intermediate complexity' dataset). The results are presented in Table 6, Table 7, and

Table 8 for the CEV, OHT, and Scree plot methods for retaining components respectively. For the T^2 statistic, the results suggest overall performance wanes as retained components decreases. As the feature subspace's dimensionality is reduced, the T^2 statistic has a smaller window to judge the process's state. Conversely, for the Q-statistic that monitors the noise subspace, the results suggest performance increases as fewer PCs are retained, resulting in a larger noise subspace.

Indeed, the pairing of the T^2 and Q-statistic for fault detection is widely adopted because of this behaviour. Johannesmeyer et al. (2002) report combined T^2 and Q-statistic performance that is higher than either of the statistic's individual performance. Having a statistic to monitor both subspaces creates a buffer effect around the optimal retained PC choice, where overall monitoring performance is largely unaffected as the retained PCs are near the optimum. The results in Table 6 and

Table 8 show a performance much better than Table 7, suggesting that the IN dataset is characterised best by eight (8) or nine (9) PCs. These results are better than what Johannesmeyer et al. (2002) report.

Table 6. The performance of patter-matching statistics using a PCA model with 9 of 14 PCs retained, chosen by the CEV method (shown in Figure 5) for the IN dataset.

	N_p	N_1	N_2	N_{db}	p (%)	η (%)	η_{max} (%)	ξ (%)
Hotelling's T^2 (95 % limit)	137	53	84	54	38.7	98.1	100	68.4
Q-statistic (95 % limit)	295	54	241	54	18.3	100	100	59.2
Hotelling's T^2 (99 % limit)	135	50	85	54	37	92.6	100	64.8
Q-statistic (99 % limit)	463	54	409	54	11.7	100	100	55.8
S_{PCA}	339	54	285	54	15.9	100	100	58

Table 7. The performance of patter-matching statistics using a PCA model with 2 of 14 PCs retained, chosen by the OHT method (shown in Figure 6) for the IN dataset.

	N_p	N_1	N_2	N_{db}	p (%)	η (%)	η_{max} (%)	ξ (%)
Hotelling's T^2 (95 % limit)	146	40	106	54	27.4	74.1	100	50.7
Q-statistic (95 % limit)	282	54	228	54	19.1	100	100	59.6

Hotelling's T^2 (99 % limit)	176	43	133	54	24.4	79.6	100	52
Q-statistic (99 % limit)	463	54	409	54	11.7	100	100	55.8
S_{PCA}	95	53	42	54	55.8	98.1	100	77

Table 8. The performance of patter-matching statistics using a PCA model with 8 of 14 PCs retained, chosen by the Scree plot method (shown in Figure 7) for the IN dataset.

	N_p	N_1	N_2	N_{db}	p (%)	η (%)	η_{max} (%)	ξ (%)
Hotelling's T^2 (95 % limit)	131	50	81	54	38.2	92.6	100	65.4
Q-statistic (95 % limit)	370	54	316	54	14.6	100	100	57.3
Hotelling's T^2 (99 % limit)	143	50	93	54	35	92.6	100	63.8
Q-statistic (99 % limit)	362	54	308	54	14.9	100	100	57.5
S_{PCA}	345	54	291	54	15.7	100	100	57.8

To investigate what the process monitoring application would reflect, a faulty run is analysed with PCA models based on the CEV and OHT method and 95 % confidence limits. Run 63 suffers from a fault that causes a step increase to the reactor's inlet flow rate (fault ID F6). The effect on the control and manipulated variables is shown in Figure 27.

For the CEV-based model, the T^2 statistic's monitoring results are shown in Figure 13 and Q-statistic's results in Figure 14. At the fault incidence, T^2 reacts with a clear step increase from below the confidence limit to above it, with a majority of the contributions from the reactor inlet flow rate variable. The Q-statistic shows a similar but more subtle change after the fault incidence. This suggests that the 9 retained PCs appropriately capture the effects of the reactor inlet flow rate on the process.

For the OHT-based model, the T^2 statistic's and Q-statistic's results are shown in Figure 15 and Figure 16, respectively. The behaviour of the statistics is now inverted when compared to the CEV-based model. The Q-statistic now shows a step increase to above it's confidence limit at the fault incidence, while the T^2 statistic's response is more muted and has almost no noise, showing only delta-dirac responses at and after the fault occurs. The contribution of the reactor inlet flow rate variable to the T^2 statistic is one of the smallest, while it is the overwhelming contributor to the Q statistic. This suggests that the two (2) retained PCs do not successfully capture the effect of the reactor inlet flowrate on the process, as it is a significant contributor in the noise subspace and not the feature subspace.

The difference in monitoring statistic behaviour between the two levels of PCA model complexity, governed by retained PCs, reaffirms the known behaviour of the T^2 and Q-statistic monitoring the separate feature and noise subspaces, respectively. This serves as an example of why they are almost always implemented together. Additionally, it supports the earlier claim that eight (8) to nine (9) PCs are required to capture the process interaction.

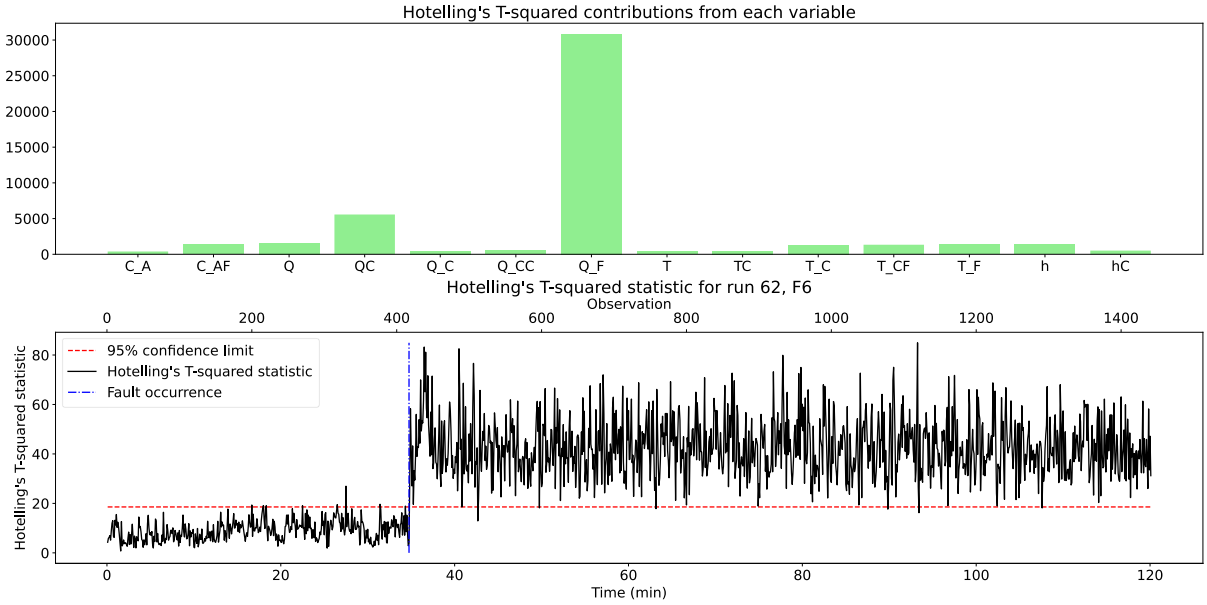


Figure 13. The T^2 monitoring results for the CEV-based PCA model when a step increase to the feed flowrate is incurred (fault ID F6). After the fault occurrence, 1 017 out of 1 024 measurements exceed the 95 % confidence limit (63 required for 95 % confidence limit of detection).

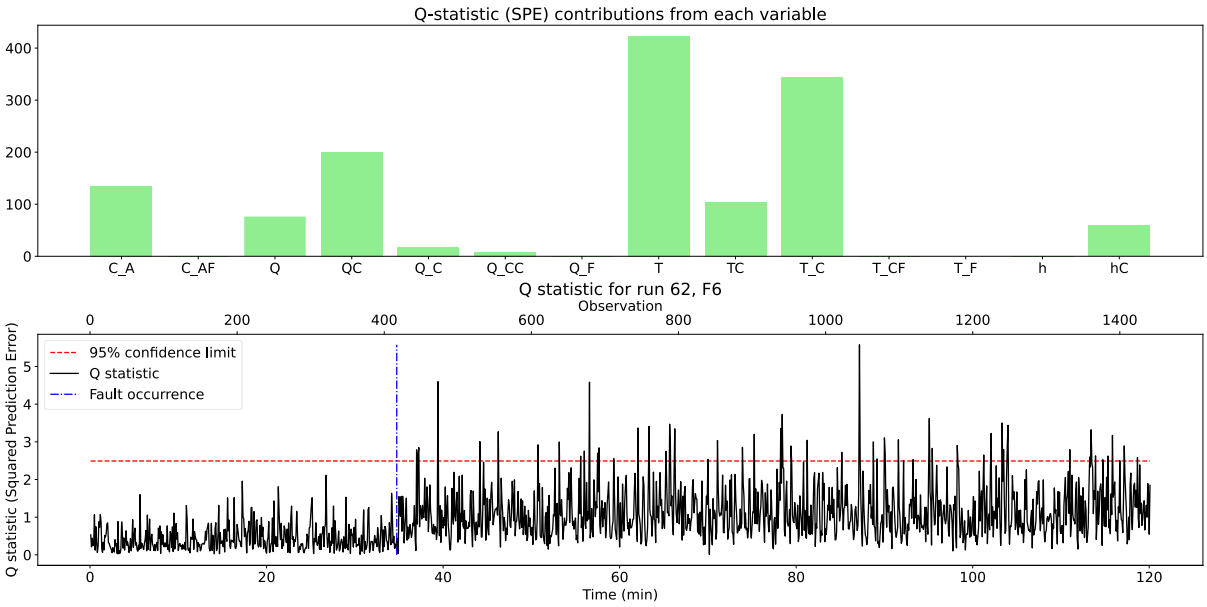


Figure 14. The Q-statistic monitoring results for the CEV-based PCA model when a step increase to the feed flowrate is incurred (fault ID F6). After the fault occurrence, 57 out of 1 024 measurements exceed the 95 % confidence limit (63 required for 95 % confidence limit of detection).

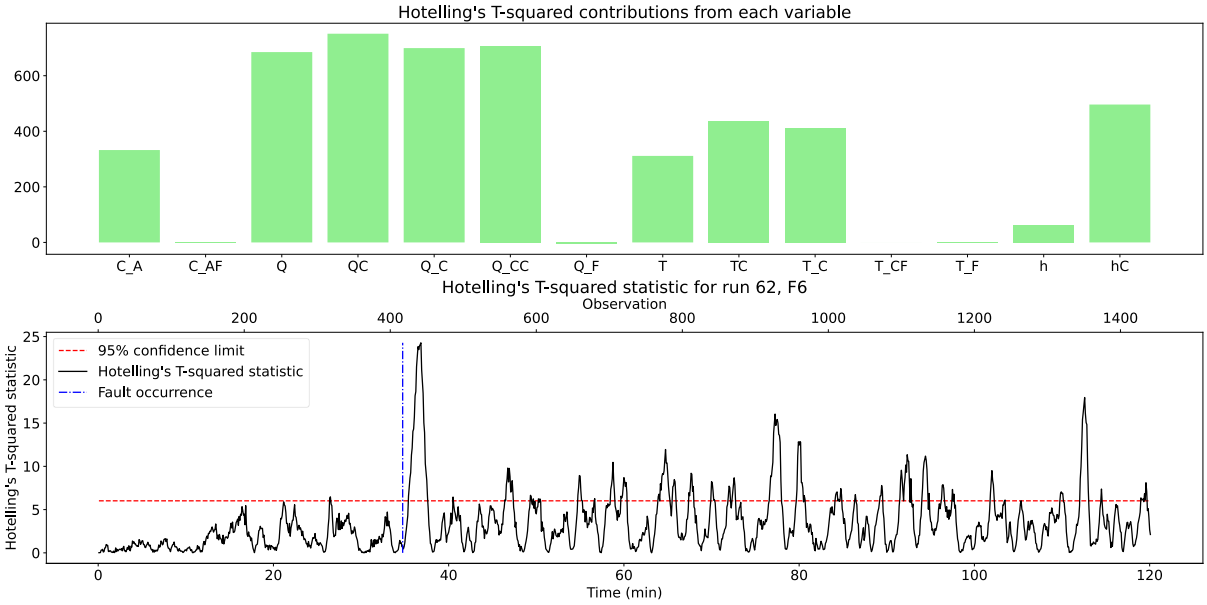


Figure 15. The T^2 monitoring results for the OHT-based PCA model when a step increase to the feed flowrate is incurred (fault ID F6). After the fault occurrence, 215 out of 1 024 measurements exceed the 95 % confidence limit (63 required for 95 % confidence limit of detection).

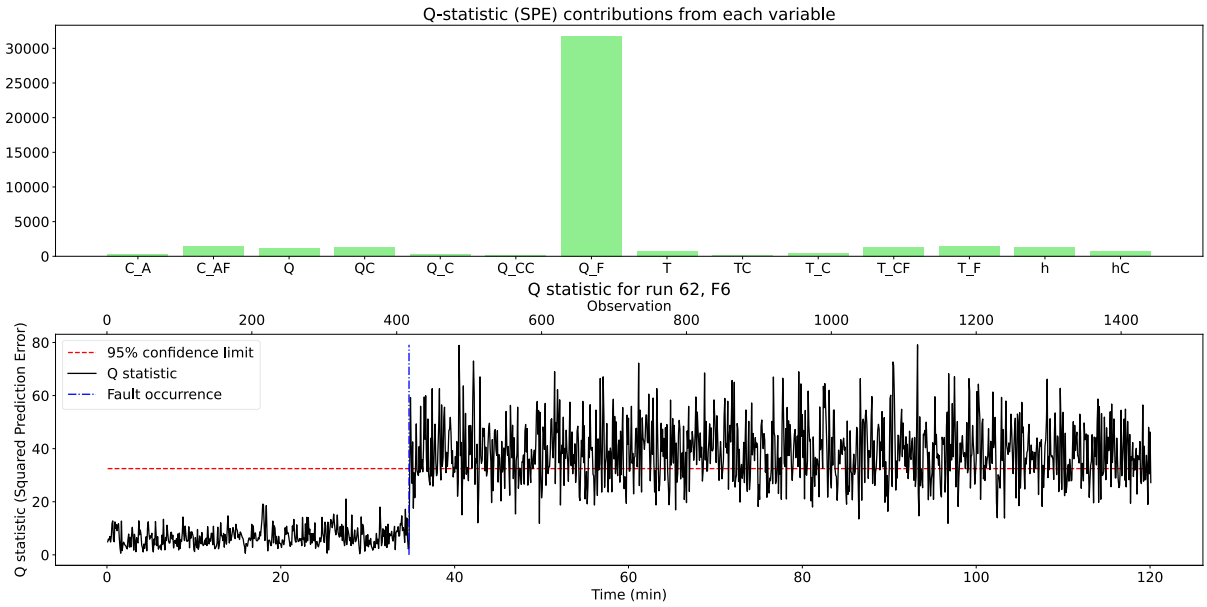


Figure 16. The Q-statistic monitoring results for the OHT-based PCA model when a step increase to the feed flowrate is incurred (fault ID F6). After the fault occurrence, 715 out of 1 024 measurements exceed the 95 % confidence limit (63 required for 95 % confidence limit of detection).

5.2.2 High noise data

For the dataset with control and both process and measurement noise, the fault detection performance is provided in Table 9, Table 10, and Table 11 for the CEV, OHT, and Scree plot methods of determining the retained PCs. Compared to the IN dataset's results, the performance is significantly degraded and falls more in line with that reported by Johannesmeyer et al. (2002). The trends

identified for the IN datset's performance reappear for the HN dataset where the T^2 statistic's performance is negatively impacted as the retained PCs are fewer. However, the Q-statistic's performance remains largely unaffected.

Comparing the 95 % and 99 % confidence limits, the T^2 statistic's overall efficiency is unaffected for the CEV method-based PCA model and improved for the Scree method-based PCA model, while the OHT-based method suffers in overall efficiency. The small change in the T^2 statistic's pool accuracy for the PCA models with more retained PCs across the 95 % and 99 % confidence limits suggests that the PC scores deviate significantly when faults occur.

Table 9. The performance of patter-matching statistics using a PCA model with 9 of 14 PCs retained, chosen by the CEV method (shown in Figure 2) for the HN dataset.

	N_p	N_1	N_2	N_{db}	p (%)	η (%)	η_{max} (%)	ξ (%)
Hotelling's T^2 (95 % limit)	120	39	81	54	32.5	72.2	100	52.4
Q-statistic (95 % limit)	386	54	332	54	14	100	100	57
Hotelling's T^2 (99 % limit)	129	40	89	54	31	74.1	100	52.5
Q-statistic (99 % limit)	453	54	399	54	11.9	100	100	56
S_{PCA}	343	49	294	54	14.3	90.7	100	52.5

Table 10. The performance of patter-matching statistics using a PCA model with 2 of 14 PCs retained, chosen by the OHT method (shown in Figure 3) for the HN dataset.

	N_p	N_1	N_2	N_{db}	p (%)	η (%)	η_{max} (%)	ξ (%)
Hotelling's T^2 (95 % limit)	192	36	156	54	18.8	66.7	100	42.7
Q-statistic (95 % limit)	350	54	296	54	15.4	100	100	57.7
Hotelling's T^2 (99 % limit)	192	31	161	54	16.1	57.4	100	36.8
Q-statistic (99 % limit)	454	54	400	54	11.9	100	100	55.9
S_{PCA}	136	46	90	54	33.8	85.2	100	59.5

Table 11. The performance of patter-matching statistics using a PCA model with 8 of 14 PCs retained, chosen by the Scree plot method (shown in Figure 1) for the HN dataset.

	N_p	N_1	N_2	N_{db}	p (%)	η (%)	η_{max} (%)	ξ (%)
Hotelling's T^2 (95 % limit)	146	40	106	54	27.4	74.1	100	50.7
Q-statistic (95 % limit)	282	54	228	54	19.1	100	100	59.6
Hotelling's T^2 (99 % limit)	176	43	133	54	24.4	79.6	100	52
Q-statistic (99 % limit)	463	54	409	54	11.7	100	100	55.8
S_{PCA}	95	53	42	54	55.8	98.1	100	77

As before, the monitoring application is investigated. Run 98 has the system experience a step increase to the reactor inlet flow rate (fault ID F6), with the control and manipulated variables shown in Figure 28.

For the CEV-based model, Figure 17 and Figure 18 display the T^2 and Q-statistic's monitoring results respectively, while Figure 19 and Figure 20, respectively, show the OHT-based model's results.

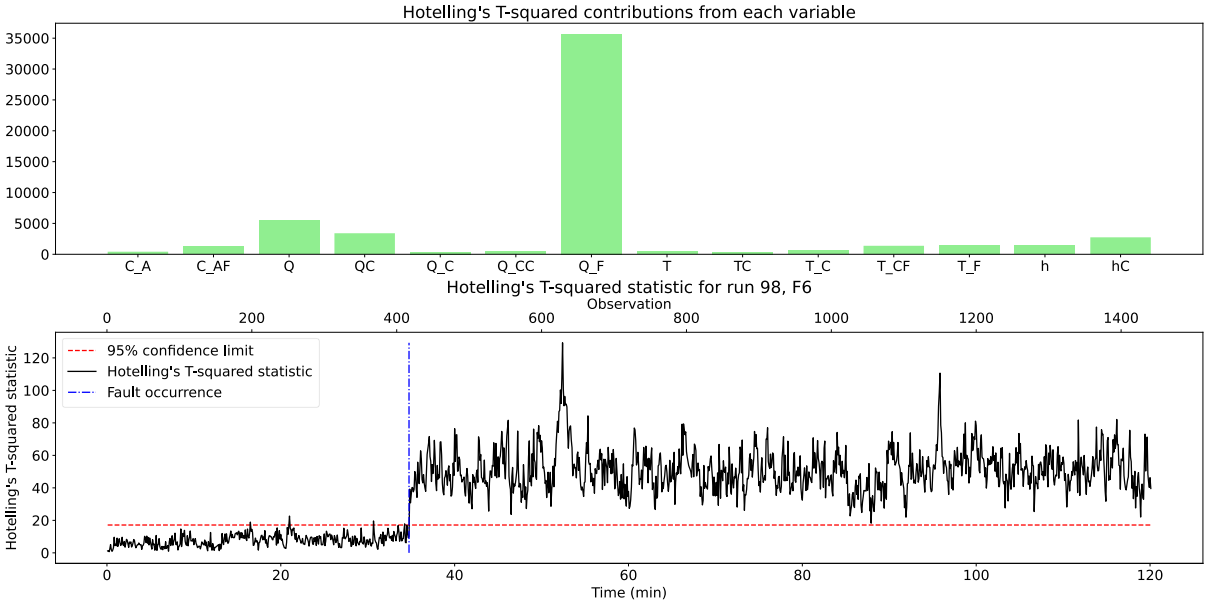


Figure 17. The T^2 monitoring results for the CEV-based PCA model when a step increase to the feed flowrate is incurred (fault ID F6). After the fault occurrence, 1 024 out of 1 024 measurements exceed the 95 % confidence limit (63 required for 95 % confidence limit of detection).

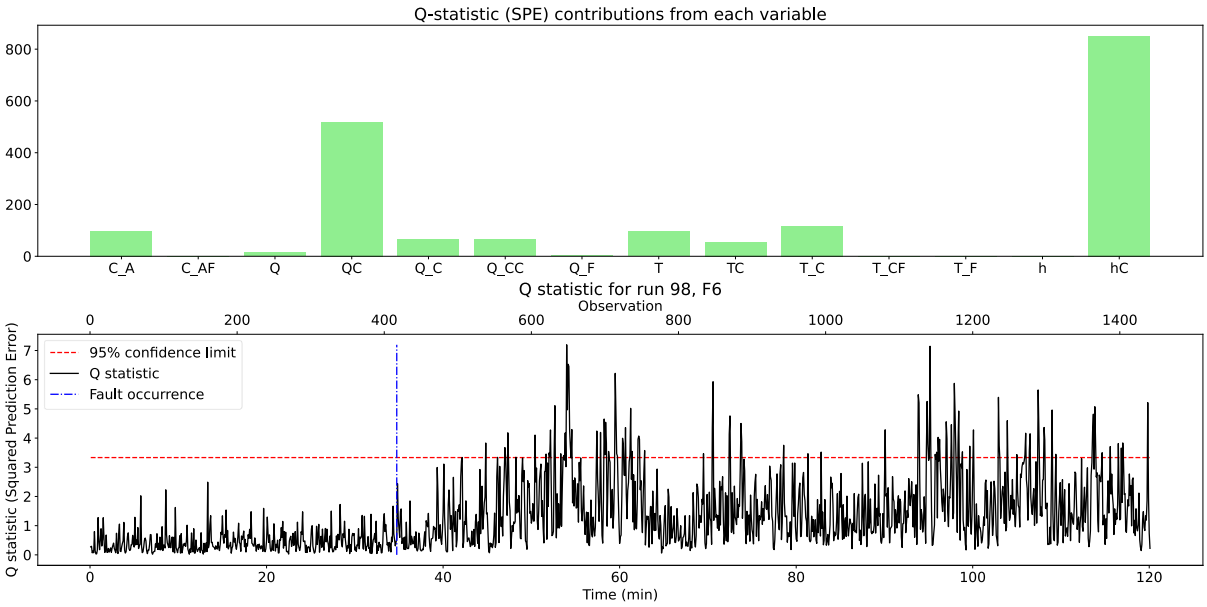


Figure 18. The Q-statistic monitoring results for the CEV-based PCA model when a step increase to the feed flowrate is incurred (fault ID F6). After the fault occurrence, 102 out of 1 024 measurements exceed the 95 % confidence limit (63 required for 95 % confidence limit of detection).

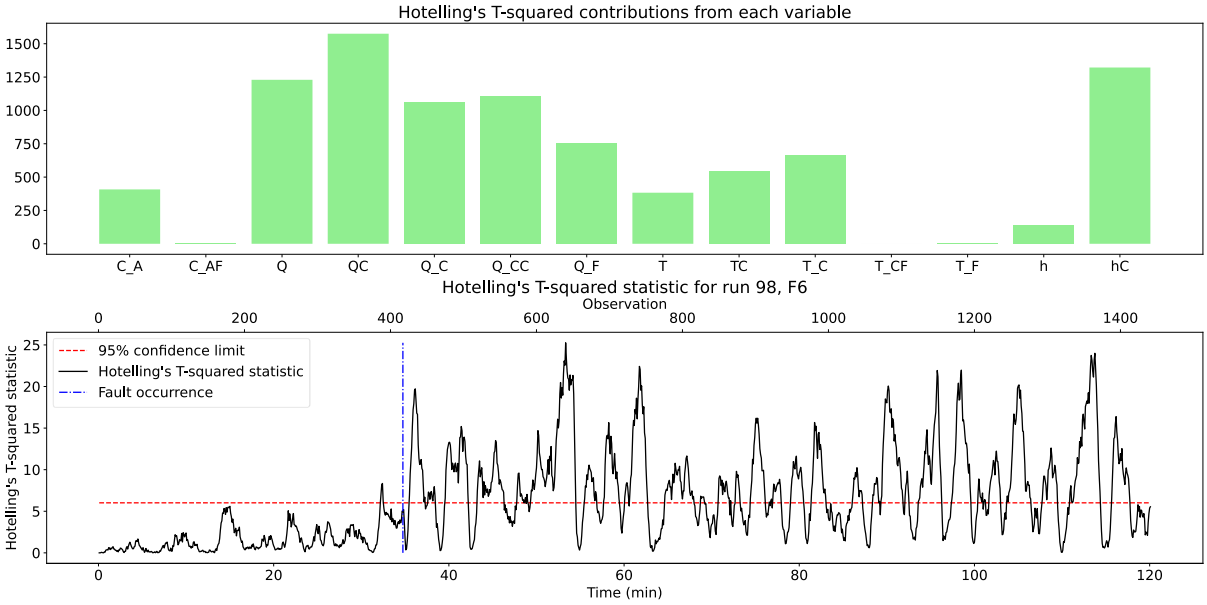


Figure 19. The T^2 monitoring results for the OHT-based PCA model when a step increase to the feed flowrate is incurred (fault ID F6). After the fault occurrence, 662 out of 1 024 measurements exceed the 95 % confidence limit (63 required for 95 % confidence limit of detection).

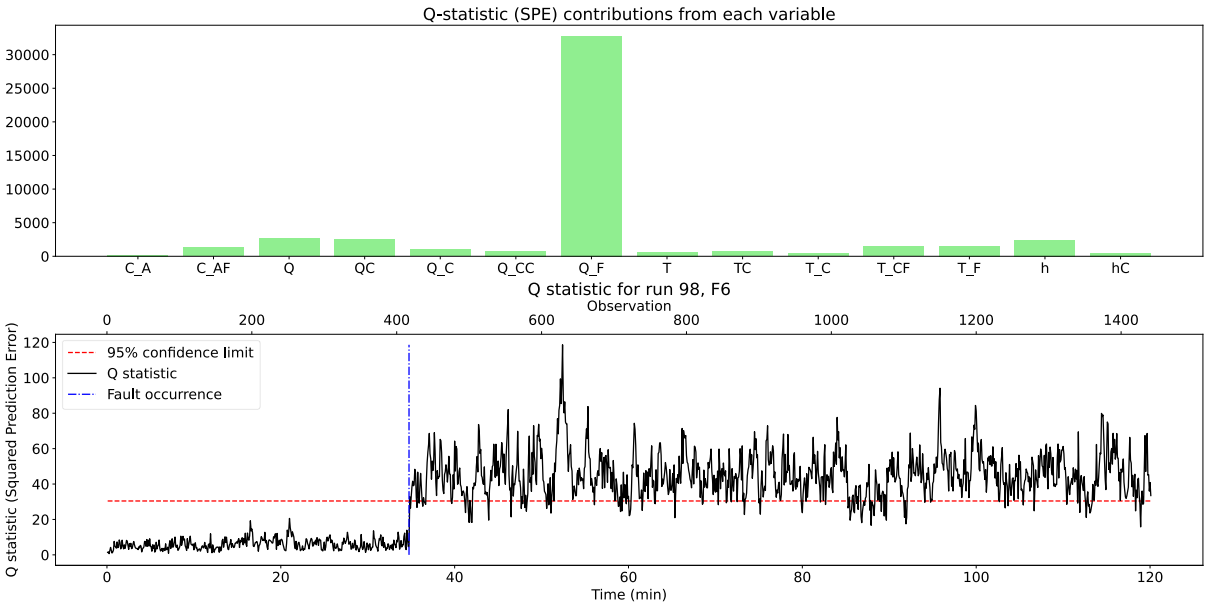


Figure 20. The Q-statistic monitoring results for the OHT-based PCA model when a step increase to the feed flowrate is incurred (fault ID F6). After the fault occurrence, 917 out of 1 024 measurements exceed the 95 % confidence limit (63 required for 95 % confidence limit of detection).

With the CEV-based model, the T^2 statistic is still able to make a clear step increase after the fault occurs in the noisier process, with an overwhelming majority contribution from the reactor inlet flow rate variable, while the Q-statistic has no clear transition at the fault time.

However, like before, the OHT-based model's Q-statistic shows a step increase to the confidence limit at the fault occurrence, while the T^2 statistic has much less noise and a transition to a much higher amplitude after the fault occurrence. This reaffirms the earlier notion of the different statistics being

responsible for monitoring their own subspace, and that the PC(s) that captures the effects of the reactor inlet flow rate is lost between eight (8) and two (2) retained PCs, but present at nine (9).

To investigate the deviation of PC scores, the scores of the first three retained PCs are plotted against one another in Figure 29 and Figure 30. However, they do not show any clear separation.

5.3 BPCA Fault Detection

Using the IN dataset, the T^2 and Q-statistic monitoring results are provided as Figure 21 and Figure 22 respectively for run 68. From Figure 21, it can be seen that the T^2 statistic is abnormally high, far above the 95 % confidence limit for the entire monitoring period, including the time of NOC. Conversely, the Q-statistic does not showcase any immediately apparent abnormality and fails to detect the faulty operation.

Strangely, the Q-statistic behaves as expected, while the T^2 does not. To investigate why, it is necessary to validate the EM algorithm used to produce the BPCA model. By ignoring the hyperparamters, ARD can be switched off and a PPCA model is acquired. MATLAB's PPCA implementation is used to validate the EM algorithm used for the BPCA model using less complex test data.

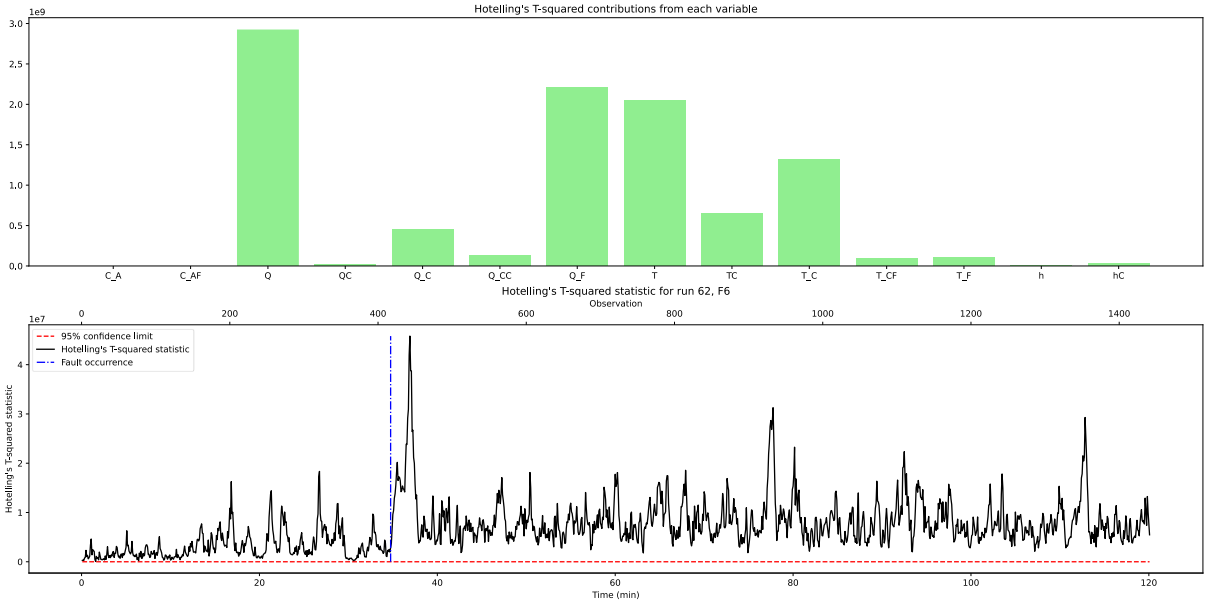


Figure 21. The T^2 monitoring results when a step increase to the feed flowrate is incurred (fault ID F6). After the fault occurrence, 1024 out of 1 024 measurements exceed the 95 % confidence limit.

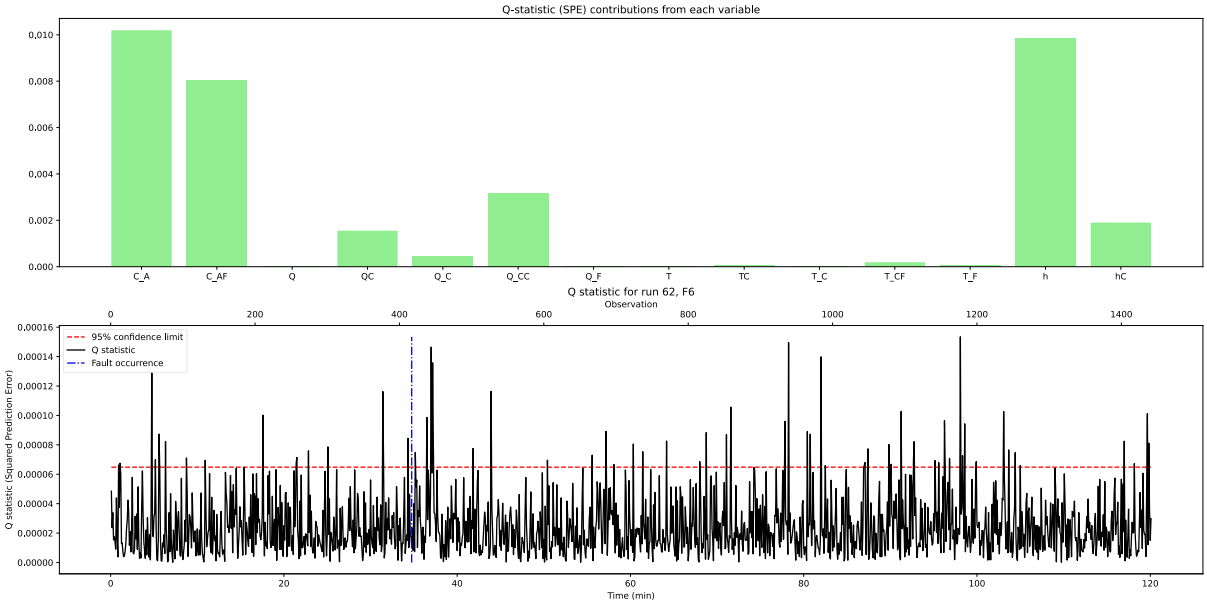


Figure 22. The Q-statistic monitoring results when a step increase to the feed flowrate is incurred (fault ID F6). After the fault occurrence, 49 out of 1 024 measurements exceed the 95 % confidence limit.

The test data is the same and Bishop's (1998) quintessential Hinton plot example, using a matrix consisting of observations across seven (7) dimensions of low variance and three (3) dimensions of high variance. The comparison is plotted in Figure 23 plotting the T^2 values of the test data the models were trained on, where the MATLAB PPCA model produces a precision of 20.81 and the BPCA EM without ARD produces a beta of 20.88. This shows the models agree and that the problem lies in the calculation of the T^2 statistic.

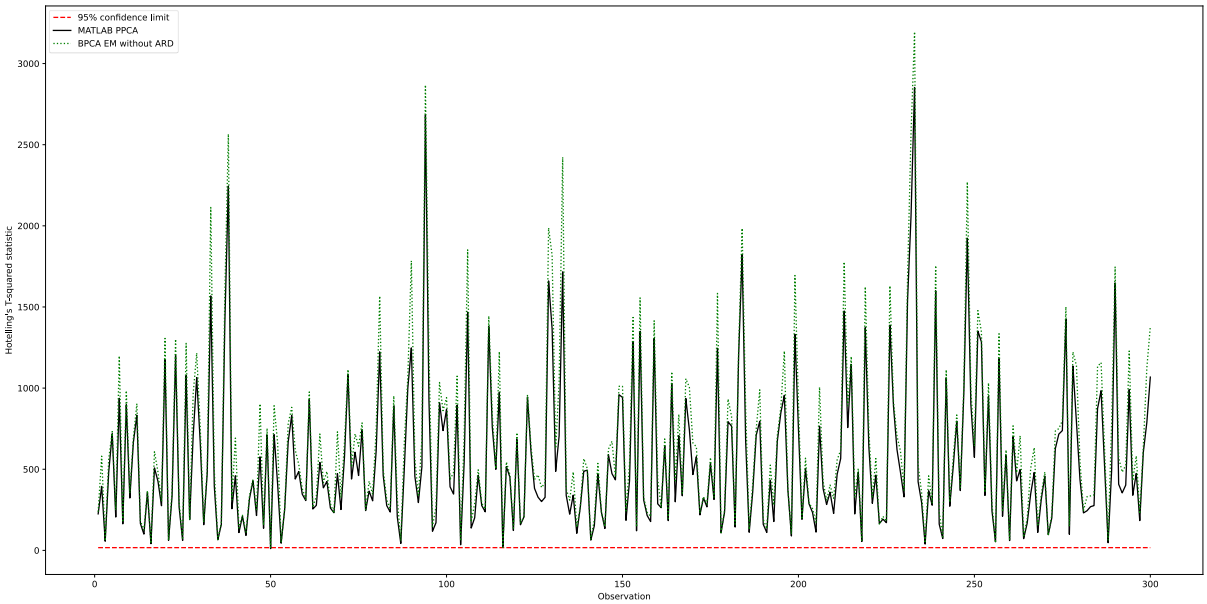


Figure 23. Validation of the BPCA EM algorithm with ARD ignored by comparison to the MATLAB PPCA implementation using test data with three (3) dimensions with high variance and seven (7) with low variance.

The Q-statistic behaving as expected suggests that the source of the T^2 values being incorrect lies in the variance used, and not the score/latent variables' values. If the scores were calculated wrong, then their analogy of the errors would also be wrong and create a Q-statistic that does not behave as expected. However, if the variance used is wrong and is that of the posterior latent variable given in equation (36), then it is unclear what the possible alternative should be.

To check this, the first three retained PCs' latent variables are plotted against one another in Figure 31 and Figure 32, which shows that their values are not extreme, and are of the same order of magnitude at the PCA model's in Figure 29 and Figure 30. This also rules out the possibility that the assumption of the evidence approximation is wrong, as a simple test data set also exhibits the unexplained behaviour.

Furthermore, we can compare the BPCA model to PCA by orthonormalizing the weight matrix, shown by Figure 24 for 100 initial guesses searched. Figure 33 in Appendix C shows that the result is identical after 1 000 initial guesses were searched. These models concur, further suggesting the issue does not lie in the EM algorithm.

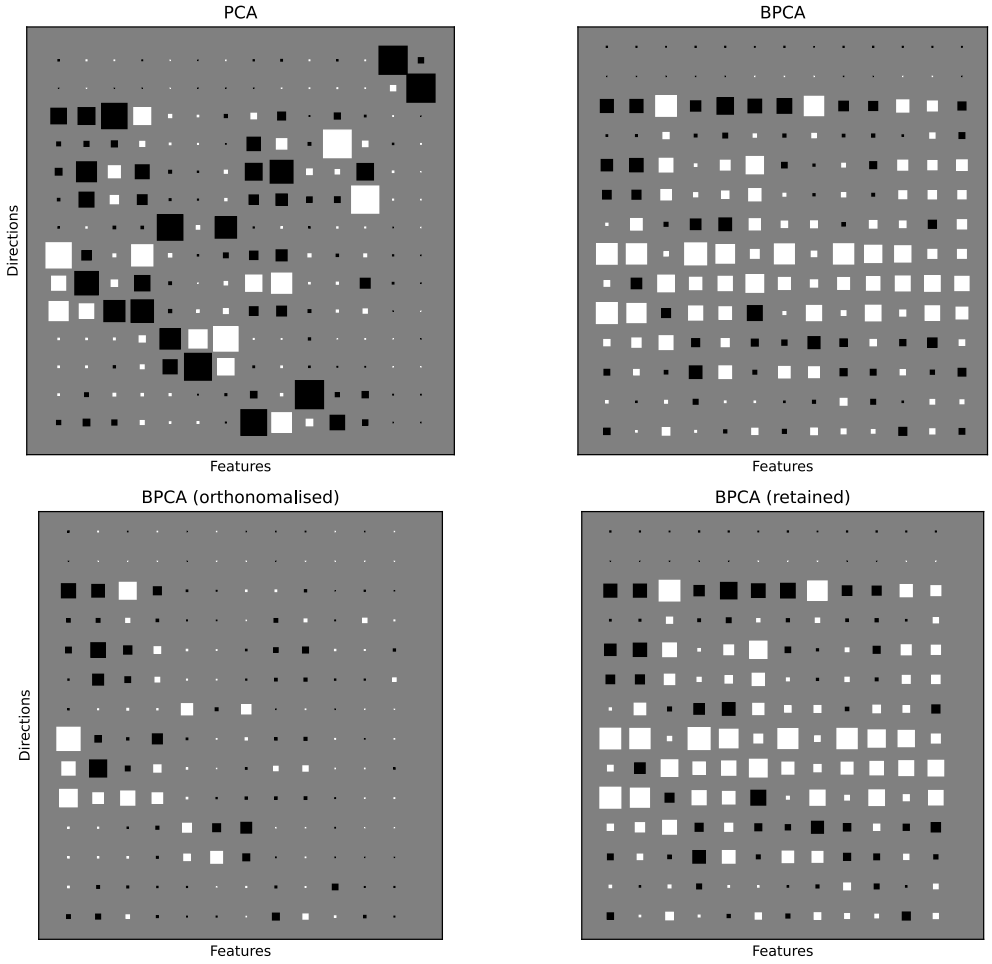


Figure 24. Hinton plots comparing the orthonormalized BPCA weight matrix against the components from standard PCA. The BPCA weight matrix is part of the EM solution that has the largest log likelihood after 100 different initial guesses were checked.

6 Conclusions

PCA is able to provide reasonable pattern-matching performance when applied to a process monitoring case study of a CSTR with varying levels of noise present in the data. PCA's performance agrees with reported performance in literature and shows the importance of pairing T^2 with the Q-statistic to monitor both the feature and noise subspaces. The PCA models are unable to create significant clusters when scores are plotted against one another, but the small effect on the T^2 statistic's pool accuracy of increasing the confidence limit from 95 % to 99 % suggests that the faulty data separates sufficiently from the NOC data. The CEV, OHT, and Scree plot methods of selecting retained PCs all provide different suggestions and the CEV method provided the best overall monitoring results, while comparison to the OHT method was able to highlight the capture of the reactor inlet flow rate's behaviour by the model is lost when retaining two (2) PCs instead of eight (8).

Unfortunately, the unexplained BPCA T^2 values prevent an in-depth comparison to the PCA results. However, the EM algorithm has been validated against MATLAB's implementation and the latent variables of the first three PC's are of the same magnitude as those from PCA. With the EM algorithm's results correct, the term most worthy of further inspection is the variance used to calculate the T^2 statistic, but an alternative to the variance of the posterior latent variable is not immediately clear.

Therefore, given the lack of native pruning of PCs and unexplained T^2 results, BPCA should not be considered for process monitoring and fault detection over standard PCA. However, it is worth investigating the causes of the T^2 results so that the computational advantage of BPCA can be used for very large industrial datasets, even though the EM algorithm takes significantly longer than PCA (in Python) for the small datasets used in this research's case study.

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Appendix B. Simulink™ Block Diagrams

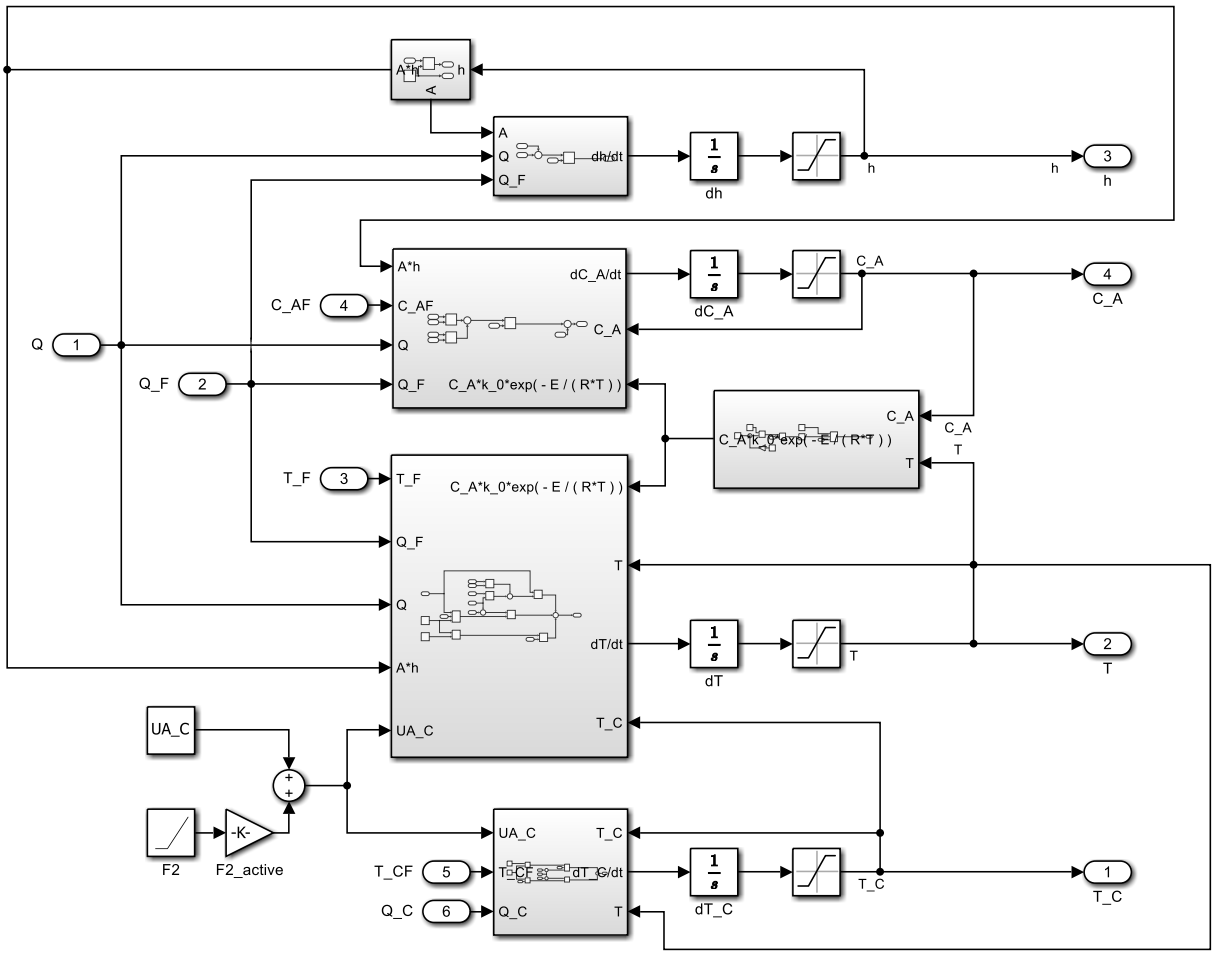


Figure 25. The CSTR Simulink™ model. Each of the four (4) major subsystems (identified by an output into an integrator block) correspond each of the conservation equations (57), (58), (59), and (60). The two (2) minor subsystems (identified by an output to another subsystem) correspond to shared terms between the two energy balances and between the one of mass and energy balances, respectively.

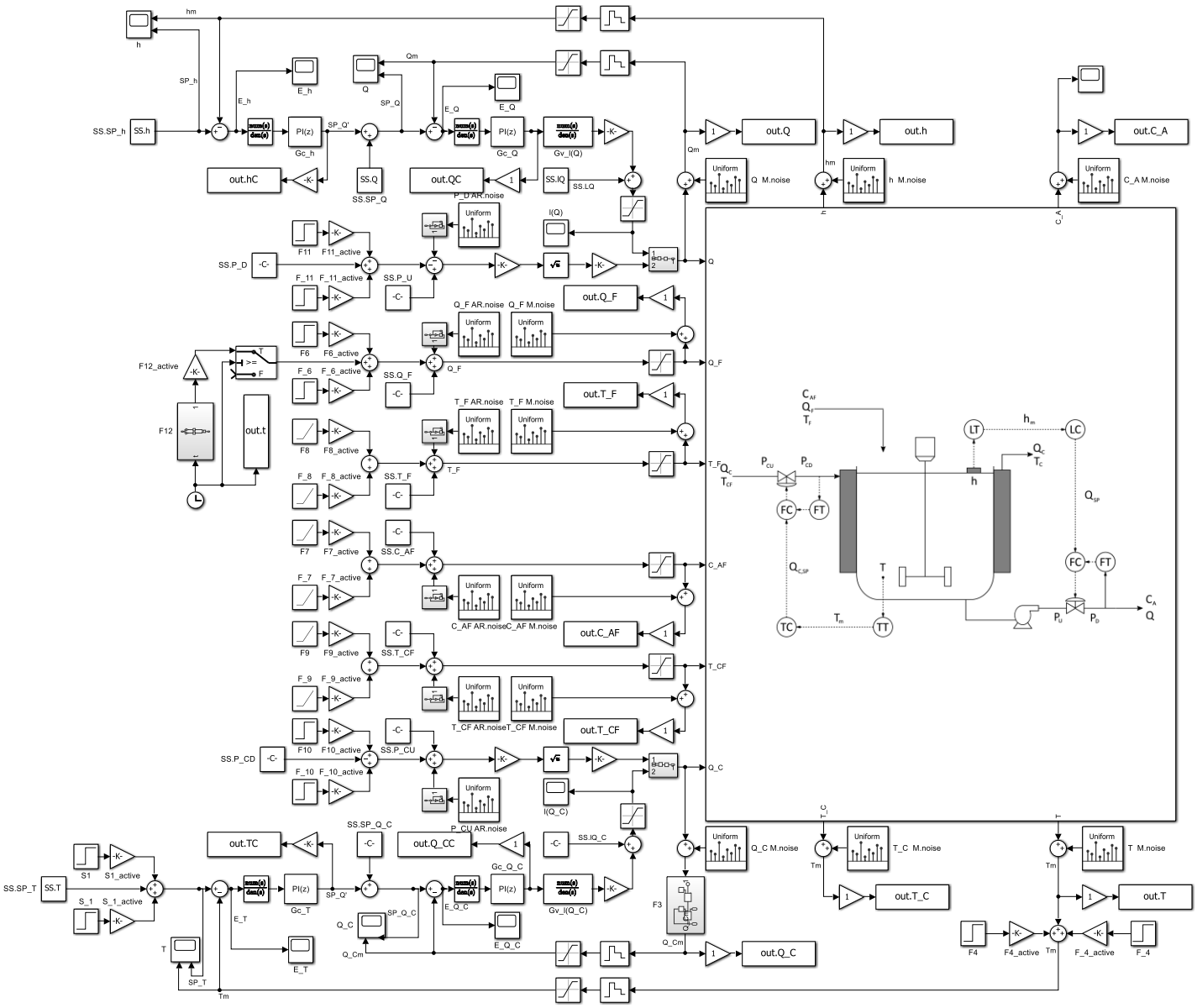


Figure 26. The CSTR Simulink™ model with control and noise. The major subsystem corresponds to the CSTR Simulink™ model shown by Figure 25.

Appendix C. Results

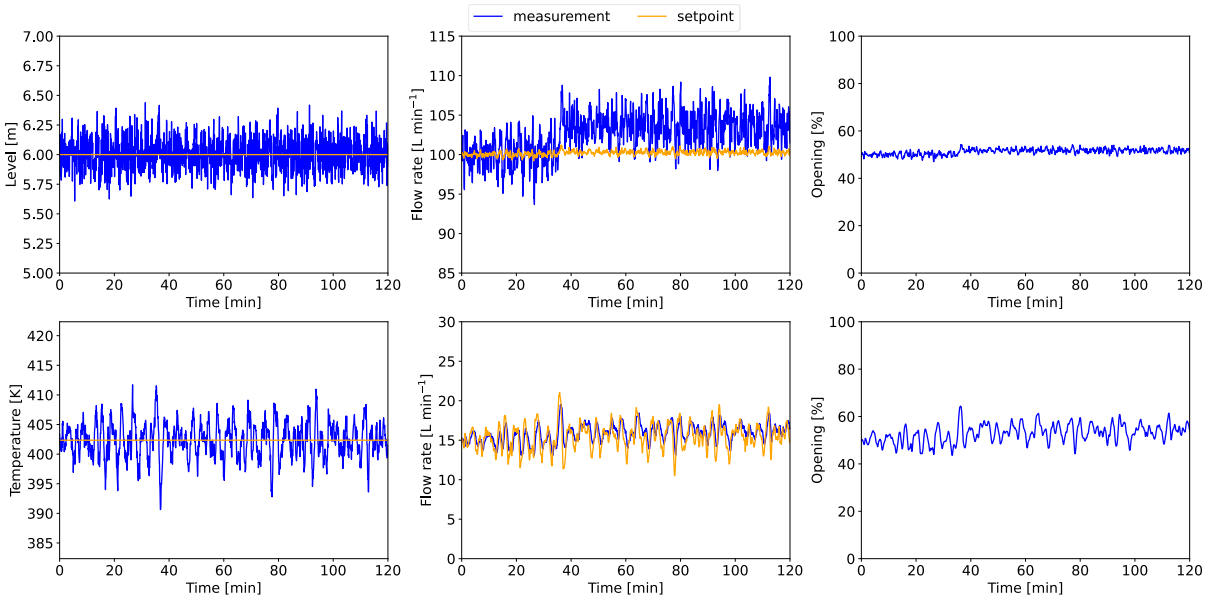


Figure 27. The control and manipulated variables' behaviour for run 62 of the IN dataset. The two rows of plots represent the two cascade control loops and their respective parameters as you progress from outer to inner loop from left to right; the primary controller's setpoint, the secondary controller's setpoint, and the final control element's position.

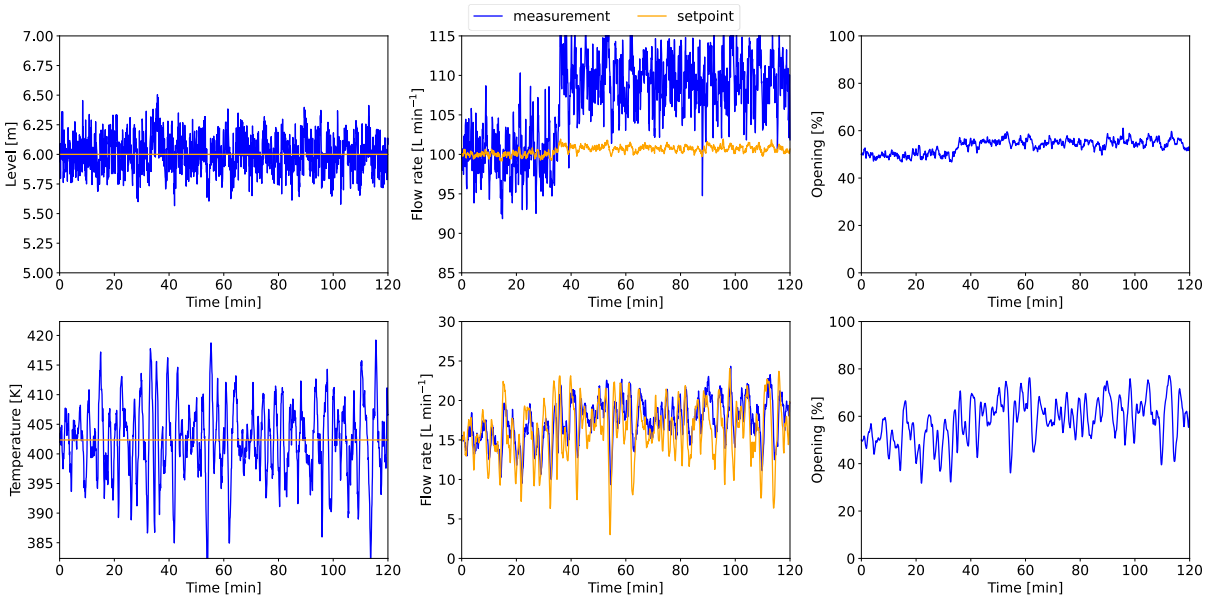


Figure 28. The control and manipulated variables' behaviour for run 98 of the HN dataset. The two rows of plots represent the two cascade control loops and their respective parameters as you progress from outer to inner loop from left to right; the primary controller's setpoint, the secondary controller's setpoint, and the final control element's position.

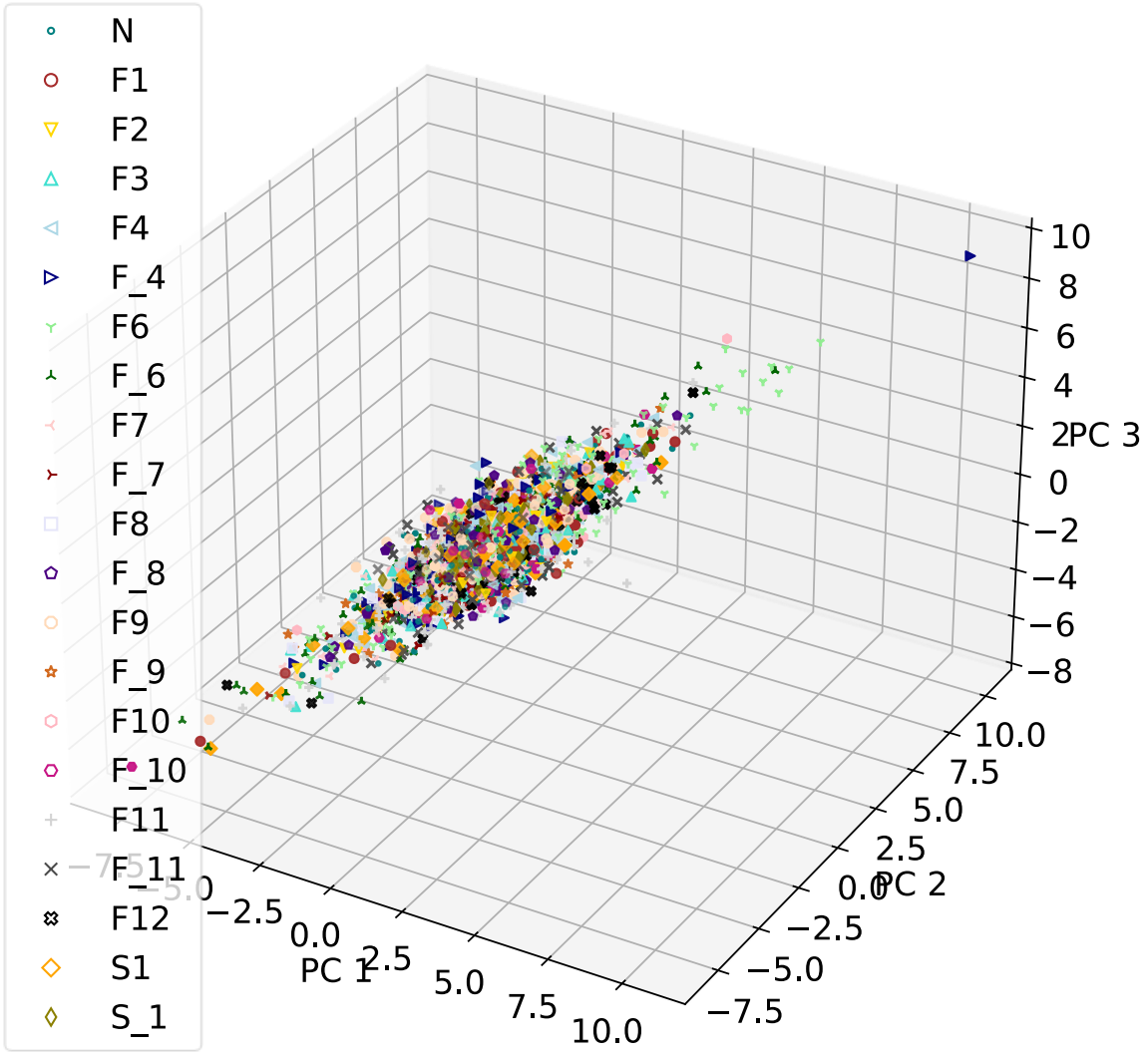


Figure 29. Three-dimensional scatter plot of the first three PC scores from every data run where the PCA model is trained on the HN dataset with retained components determined by the CEV method.

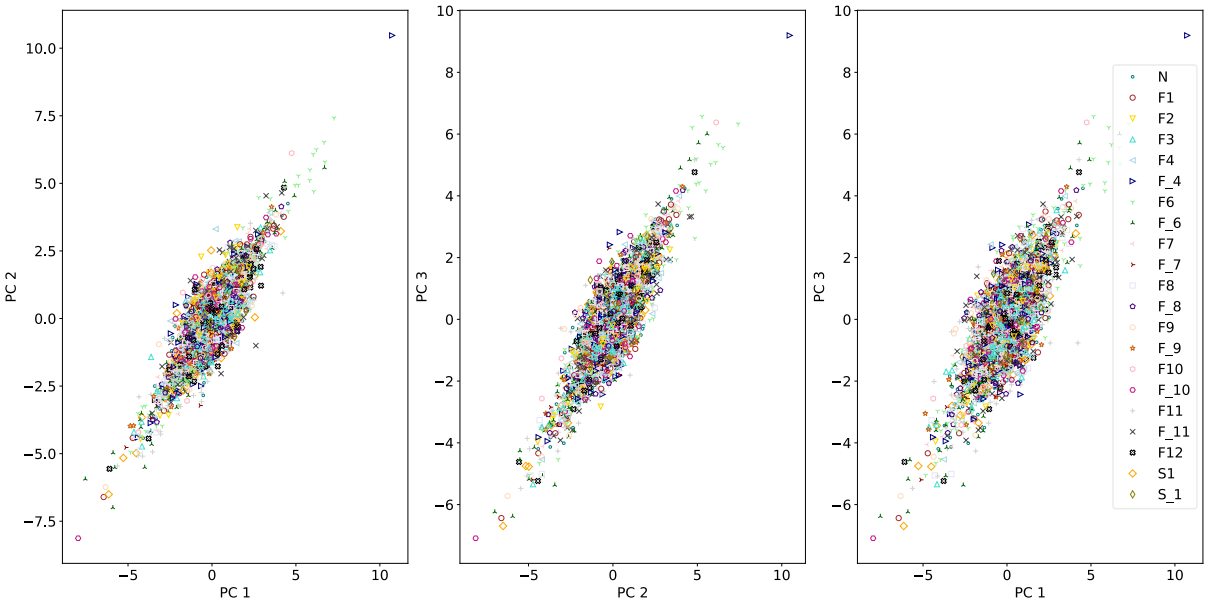


Figure 30. Side-by-side view of each combination of the first three latent variables plotted against one another. The latent variables are from every data run where the PCA model is trained on the HN dataset with retained components determined by the CEV method.

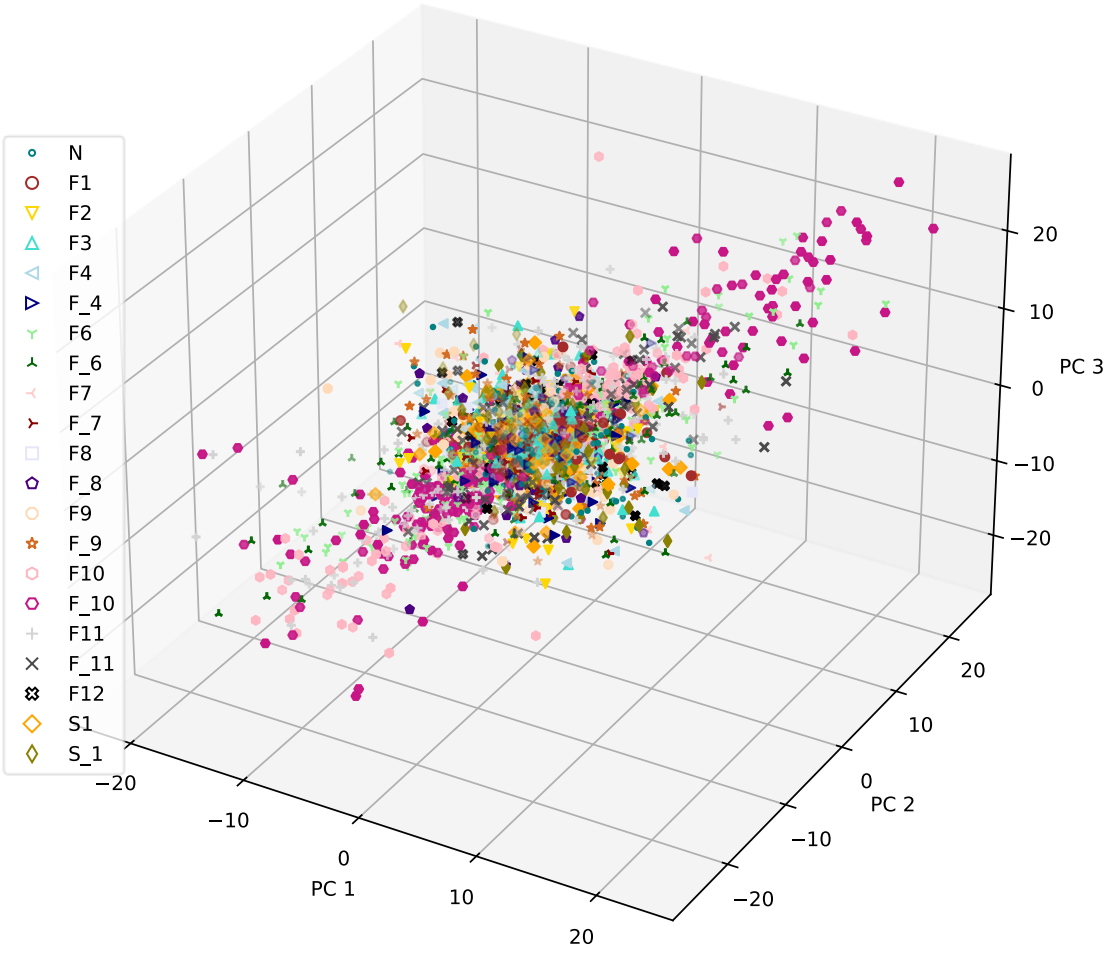


Figure 31. Three-dimensional scatter plot of the first three latent variables for every data run with the BPCA model trained on the IN dataset.

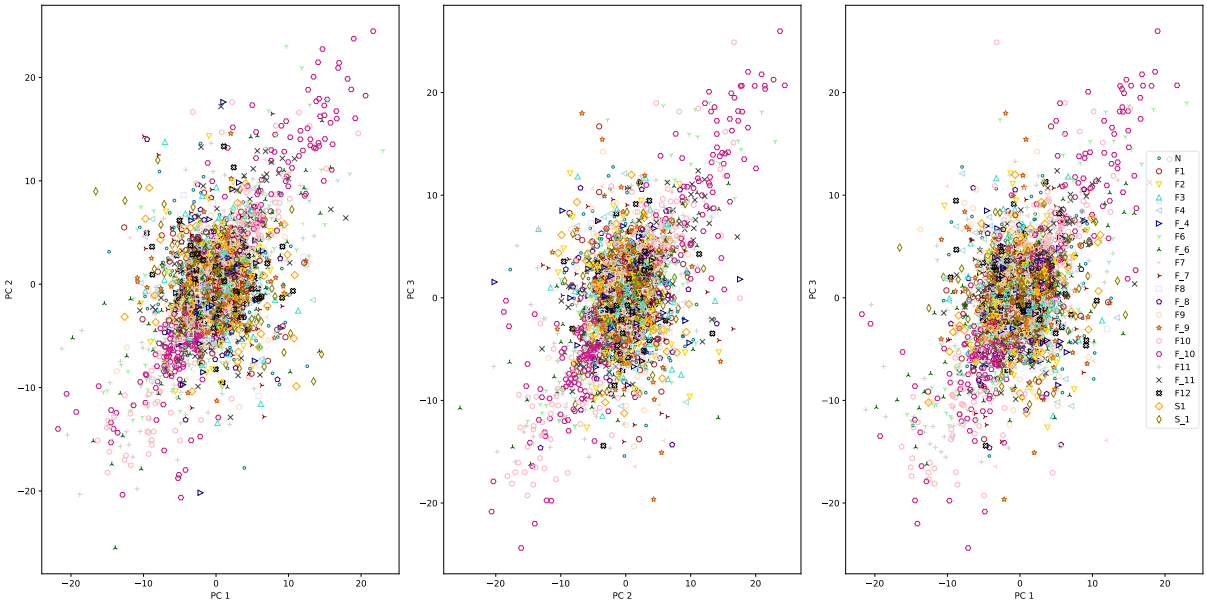


Figure 32. Side-by-side view of each combination of the first three latent variables plotted against one another. The latent variables are from every data run with the BPCA model trained on the IN dataset.

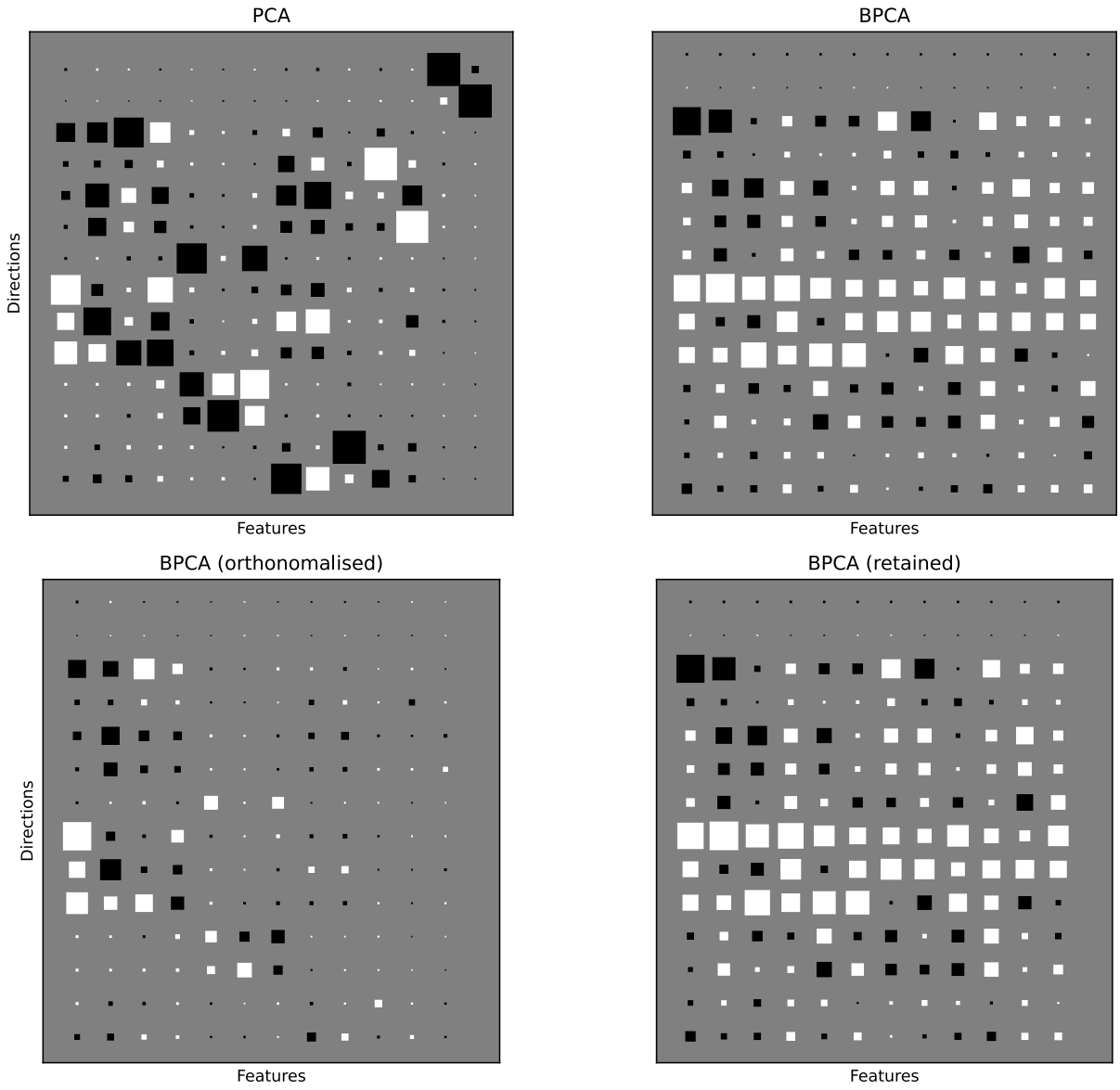


Figure 33. Hinton plots comparing the orthonormalized BPCA weight matrix against the components from standard PCA. The BPCA weight matrix is part of the EM solution that has the largest log likelihood after 1 000 different initial guesses were checked.

Appendix D. Code

All code used for this project is made available in a public GitHub repository accessible via the permalink https://github.com/mivobd/Scripsy.